

Not just a catalogue combination method, or
How to use probabilistic macauff cross-
matches to remove junk objects from your data
sample, clean your objects' measurements, find
interesting new objects, and more!

Tom J. Wilson (he/him), Tim Naylor, and Andy Wilson
t.j.wilson@exeter.ac.uk



LSST@Europe8, 30/Sep/26

 @Onoddil [@pm.me](mailto:tom@onoddil.pm.me) 
[.github.io](https://github.com/Onoddil) 



Tom J Wilson @onoddil

Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions

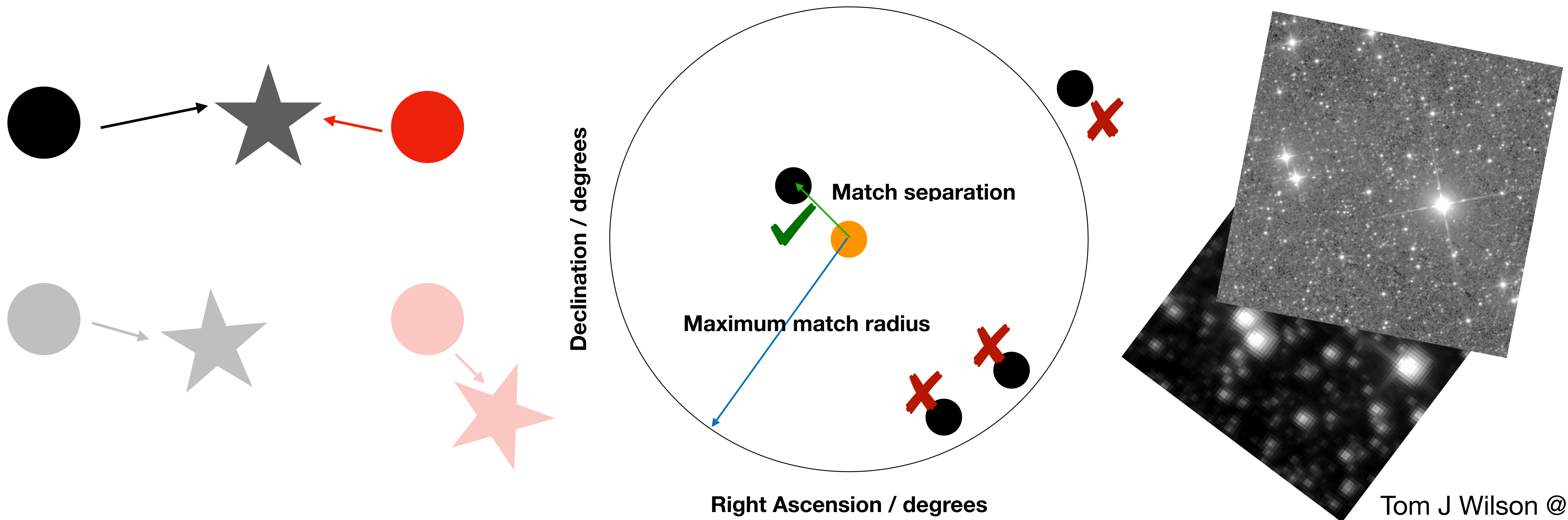


<https://github.com/macauff/macauff>

Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

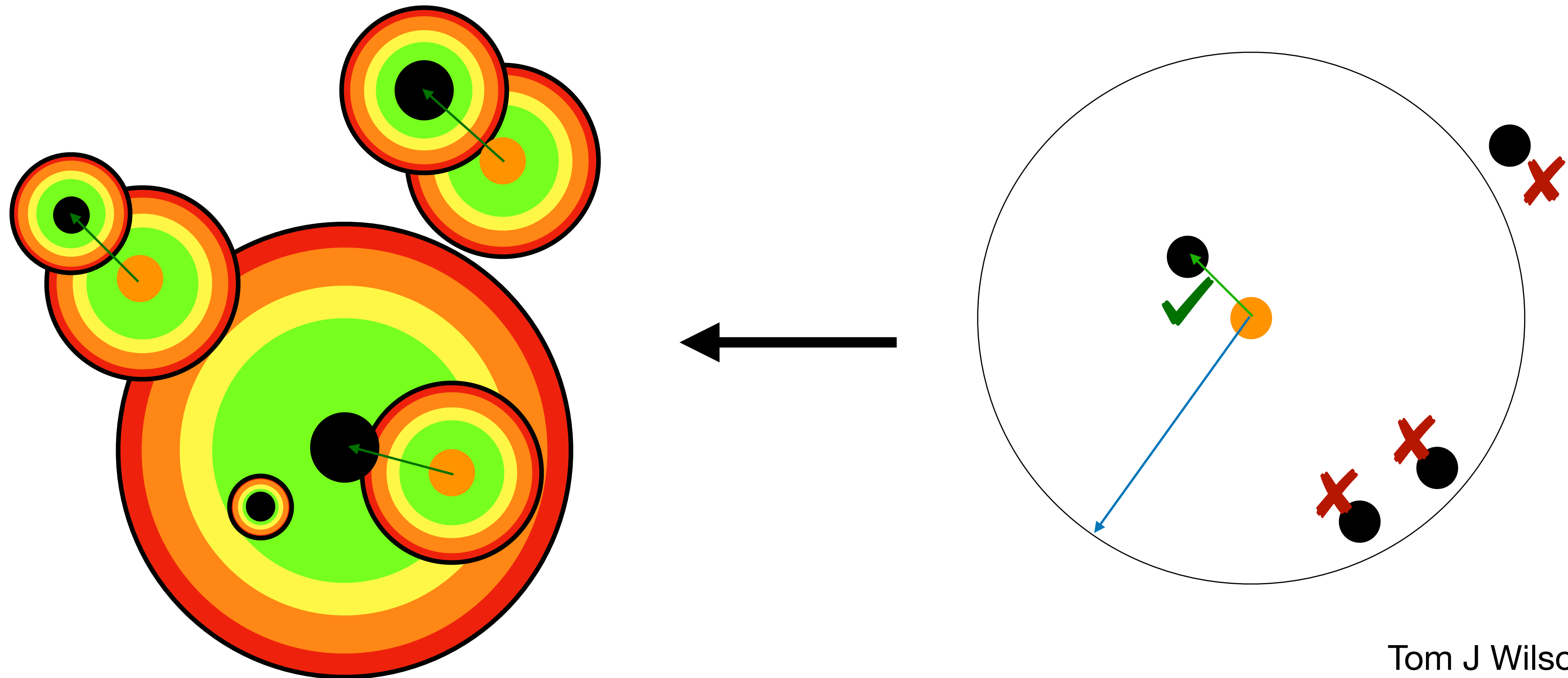
- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions



Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

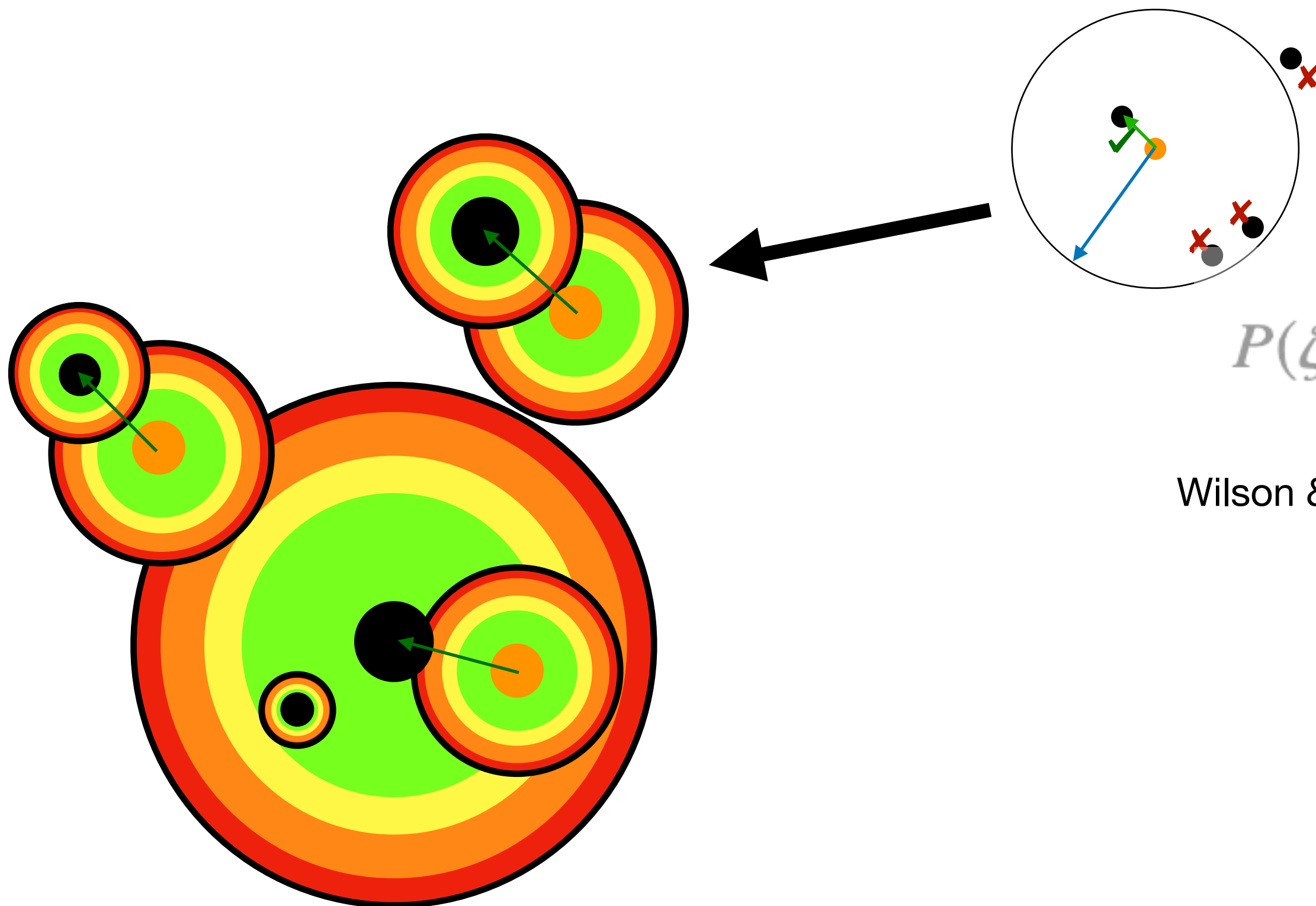
- is a **“many-to-many” algorithm**, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions



Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is **fully Bayesian**, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions



Probability of two sources having their on-sky separation given the hypothesis they are counterparts

$$P(\zeta, \lambda, k | \gamma, \phi) = \frac{1}{K} \times \prod_{\delta \notin \zeta \cap \delta \in \gamma} N_{\gamma} f_{\gamma}^{\delta} \prod_{\omega \notin \lambda \cap \omega \in \phi} N_{\phi} f_{\phi}^{\omega} \prod_{i=1}^k N_c G_{\gamma\phi}^{\zeta_i \lambda_i} c_{\gamma\phi}^{\zeta_i \lambda_i}$$

Wilson & Naylor (2018a)

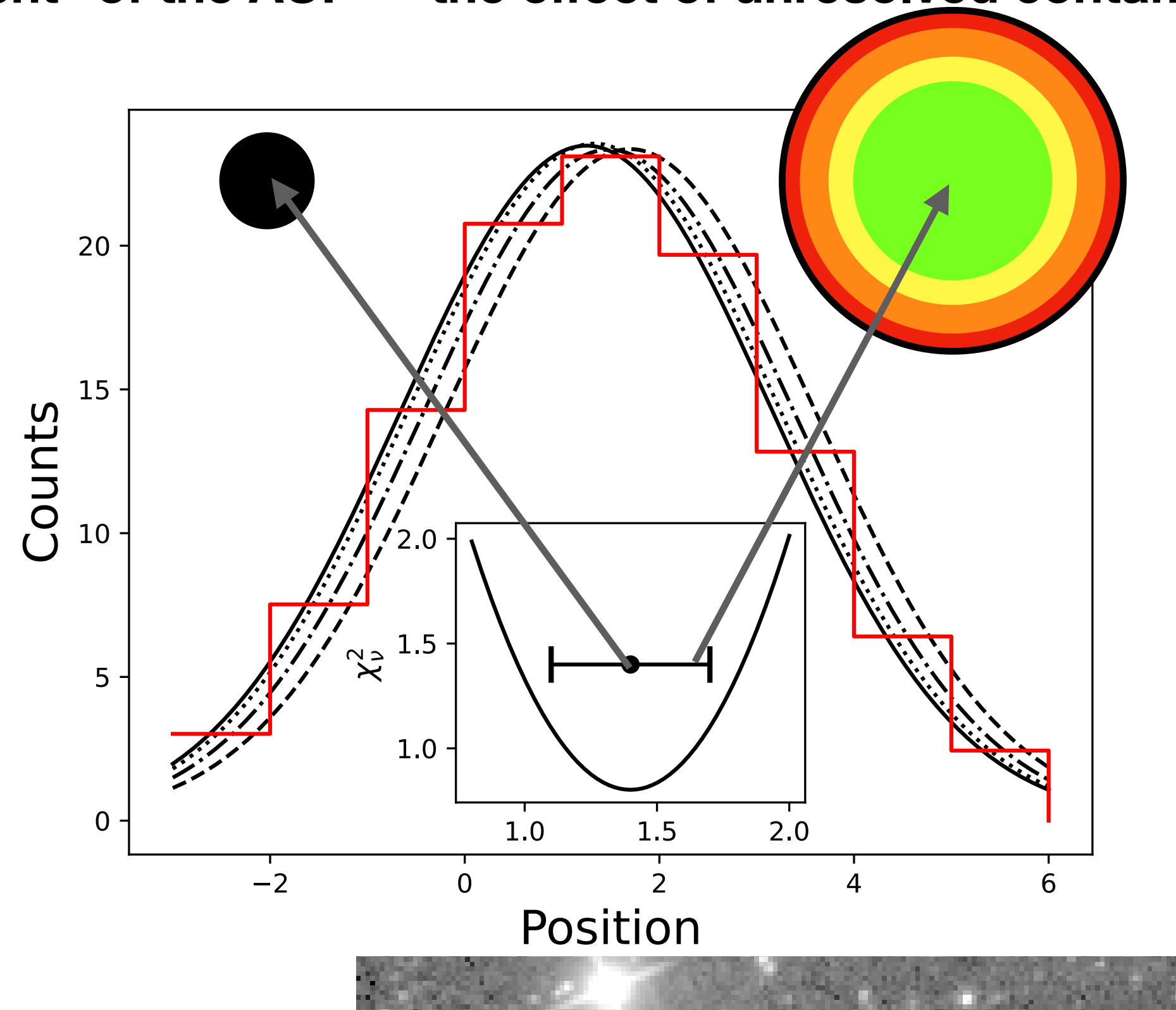
Probability of sources having their brightnesses given they are unrelated to one another (“field stars”)

Probability of sources having their brightnesses given they are counterparts

Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

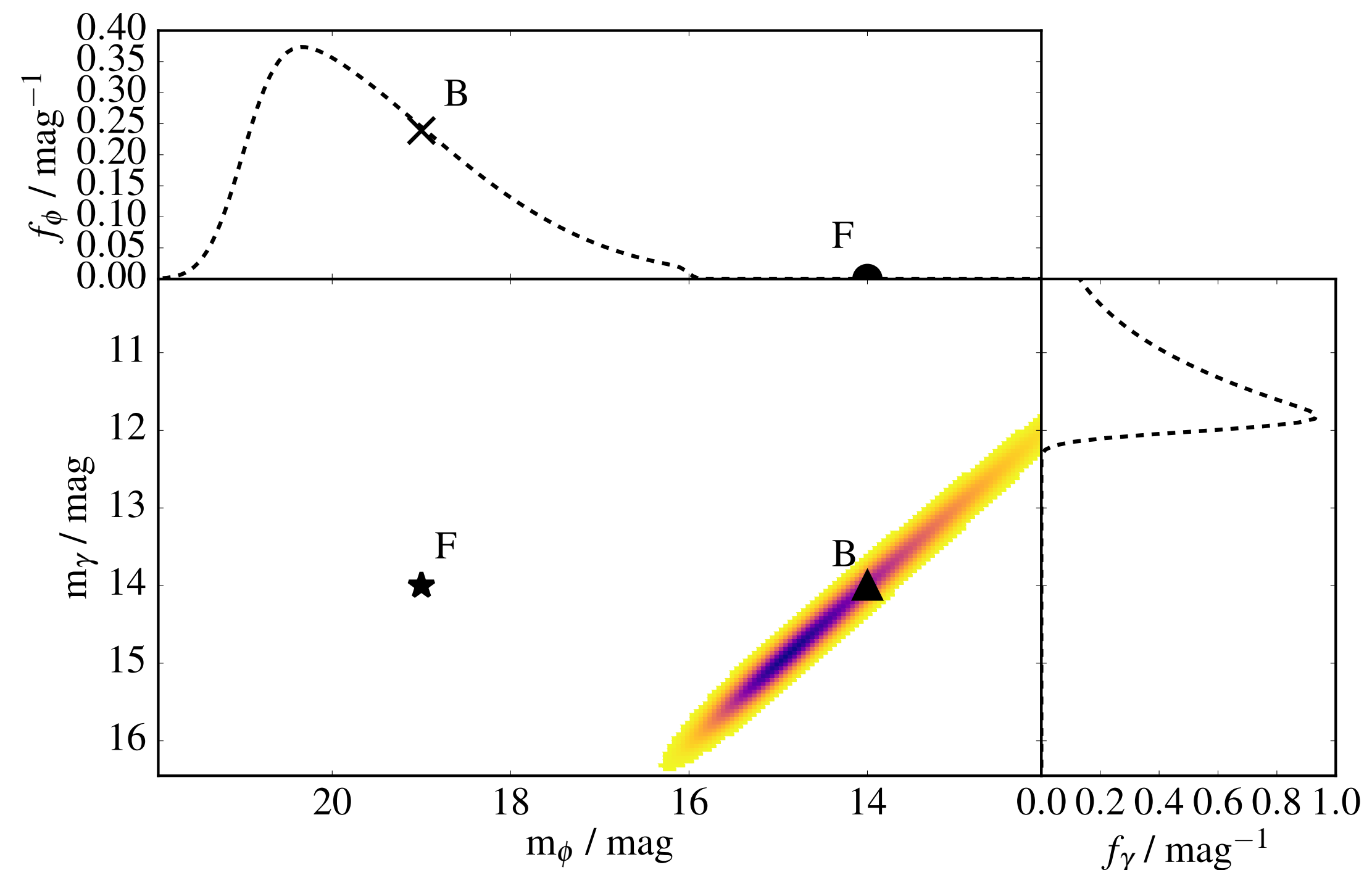
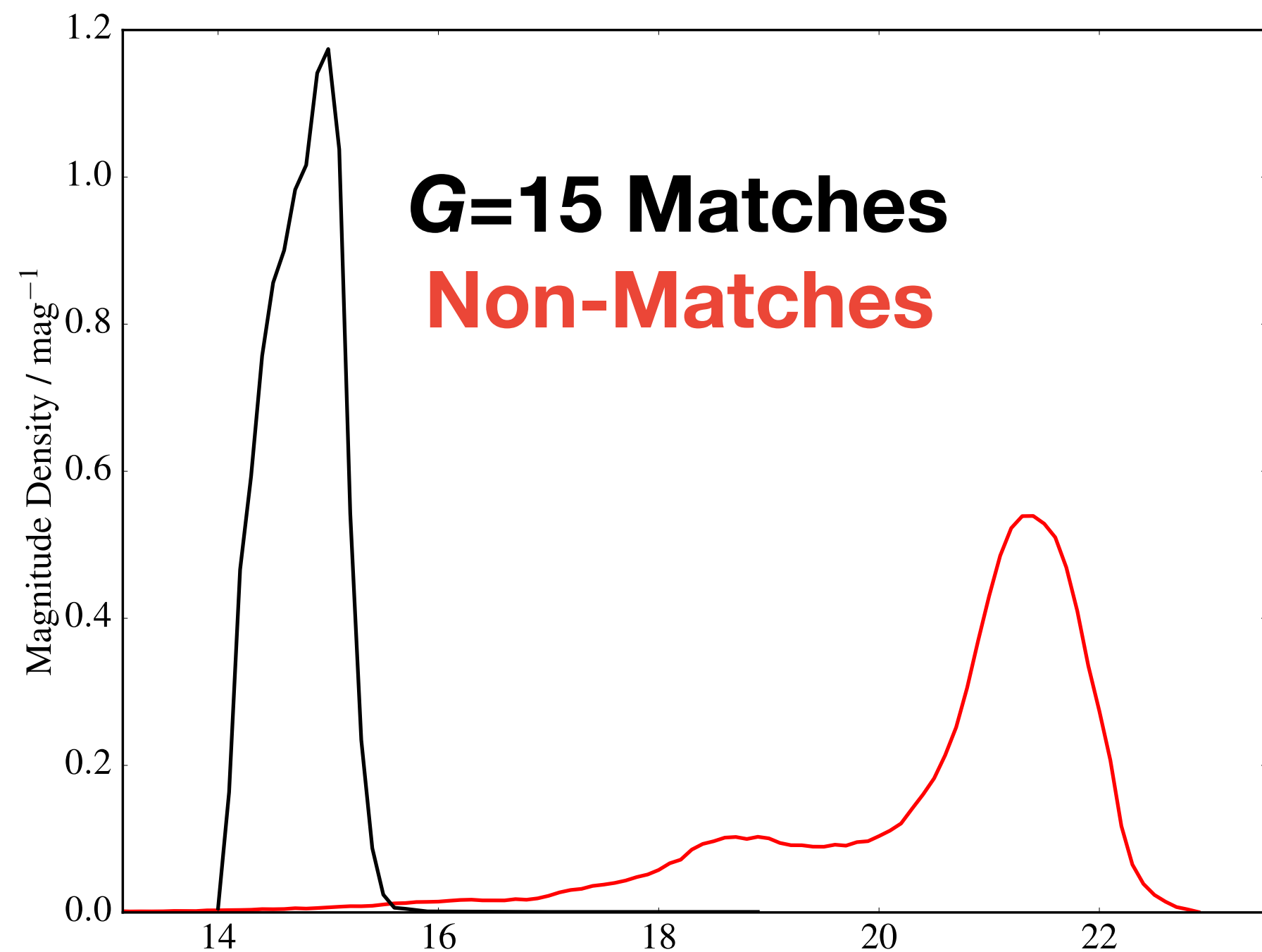
- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions



Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured **colours of potential matches** as a further discriminator of true and false matches
- extends the “Astrometric Uncertainty Function” to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions



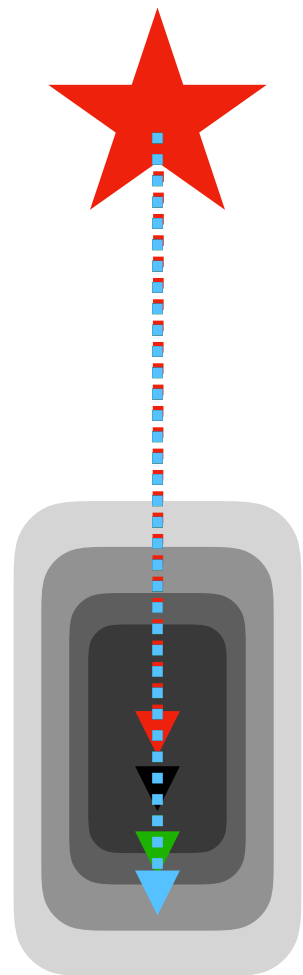
Matching Across Catalogues using the Astrometric Uncertainty Function and Flux



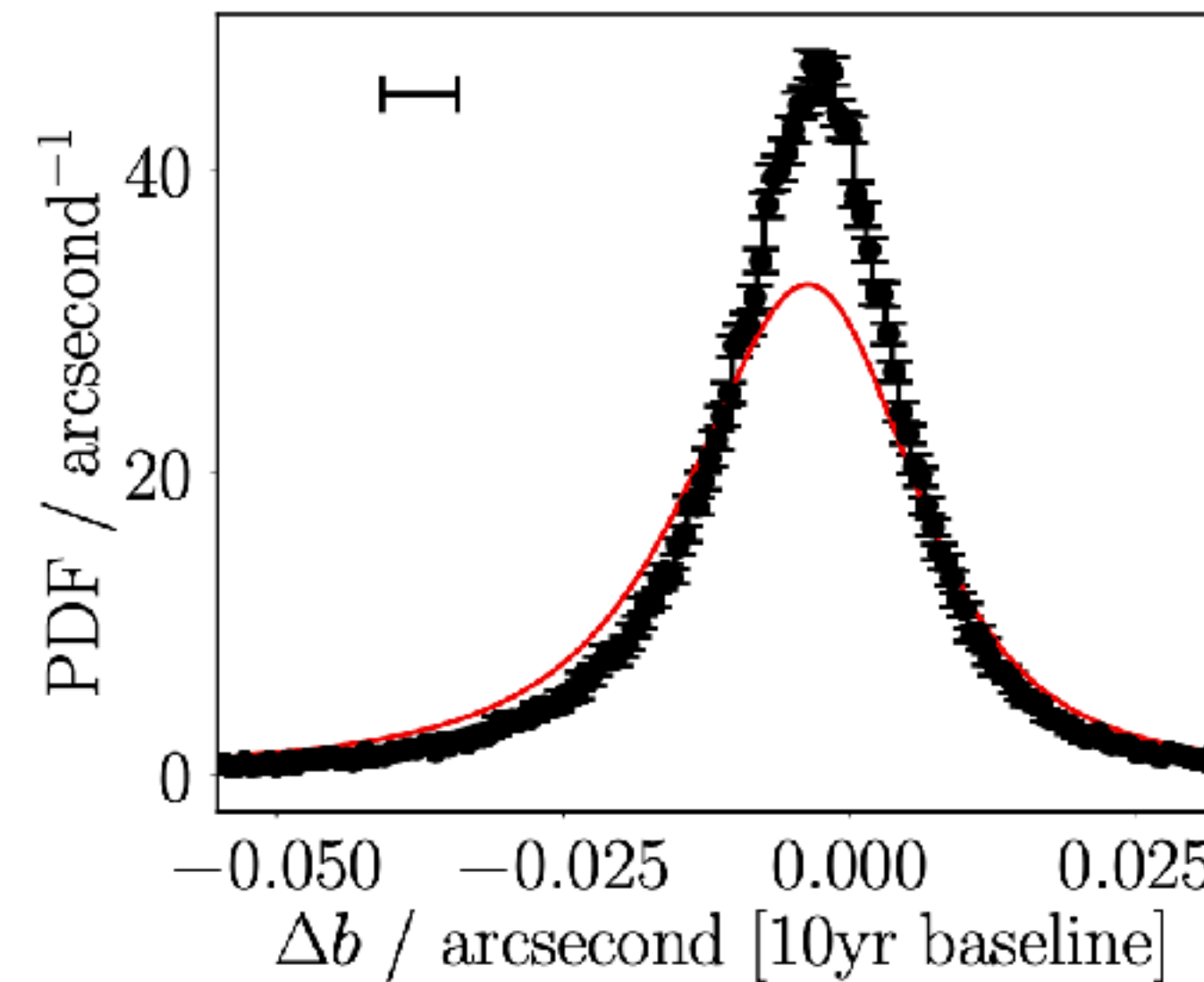
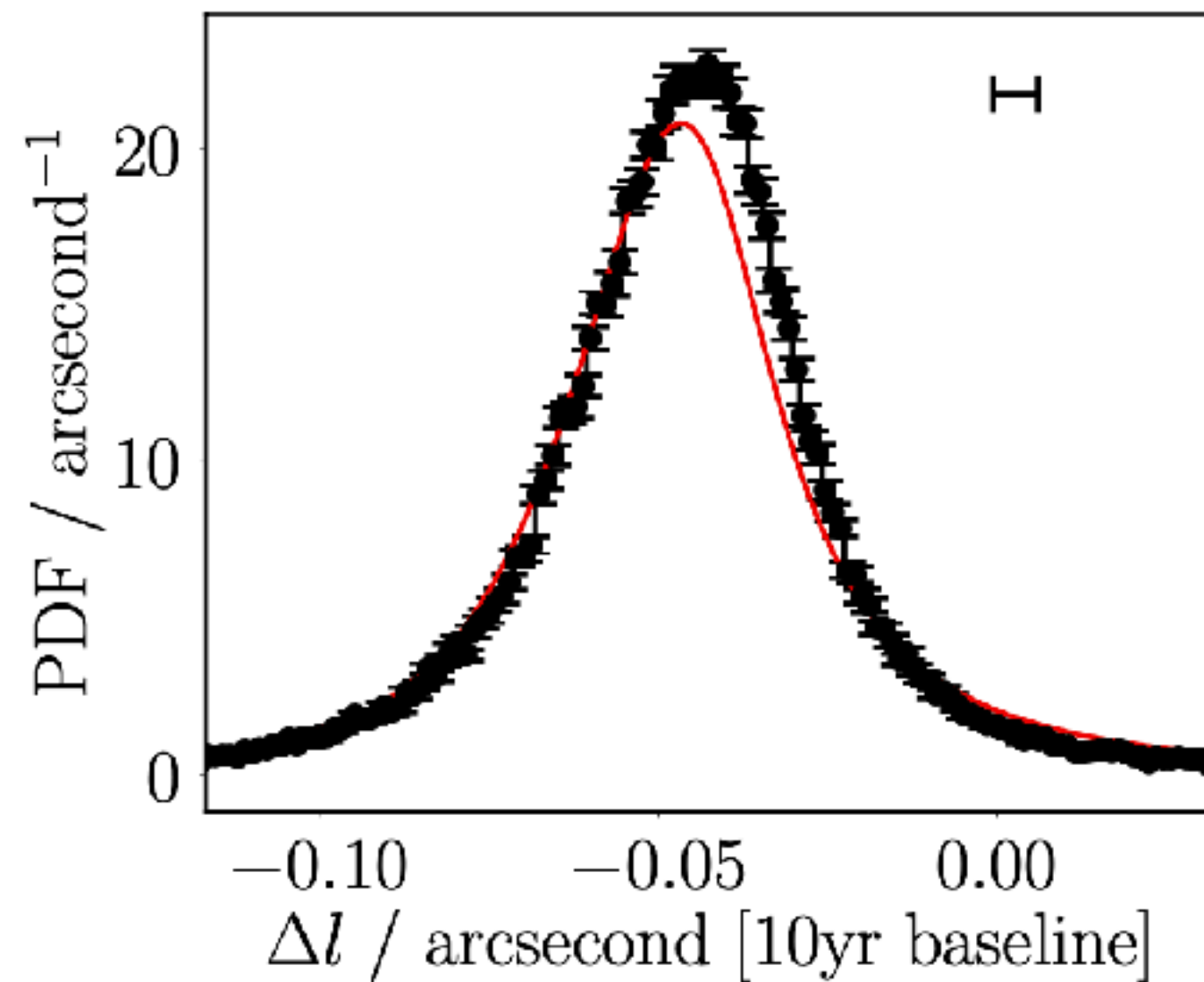
macauff is a cross-match algorithm that:

- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- **extends the “Astrometric Uncertainty Function”** to include more than just the *centroid* uncertainty, adding e.g.
 - **unknown or weakly-known proper motion** of stars
 - the “Perturbation Component” of the AUF — the effect of unresolved contaminants on measured positions

Object in 2015



Projected to 2025

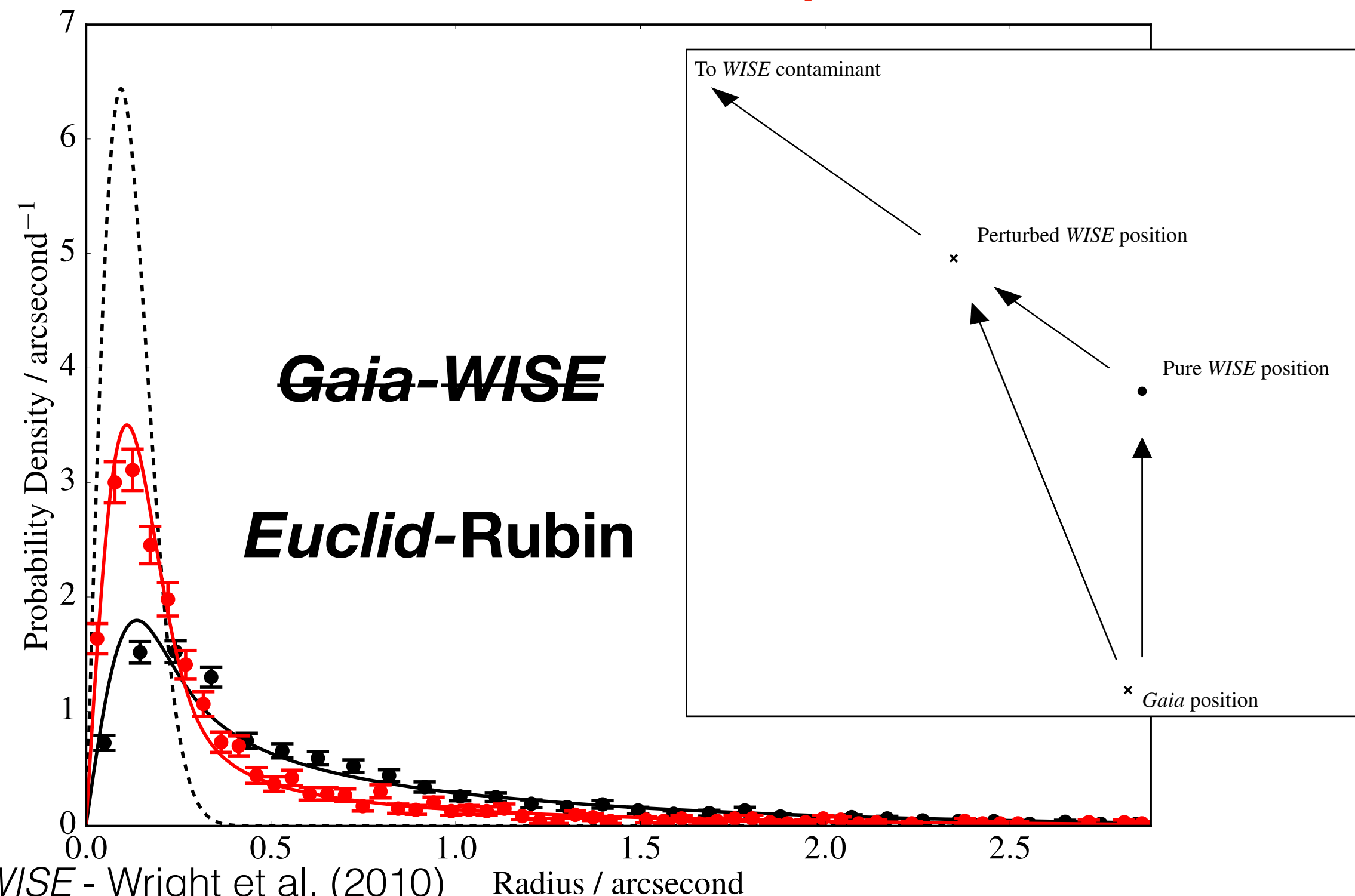


(This also applies to *uncertain* proper motions, where we can incorporate the covariance matrix of weakly-constrained proper motions, e.g. just above the single-visit LSST limit)

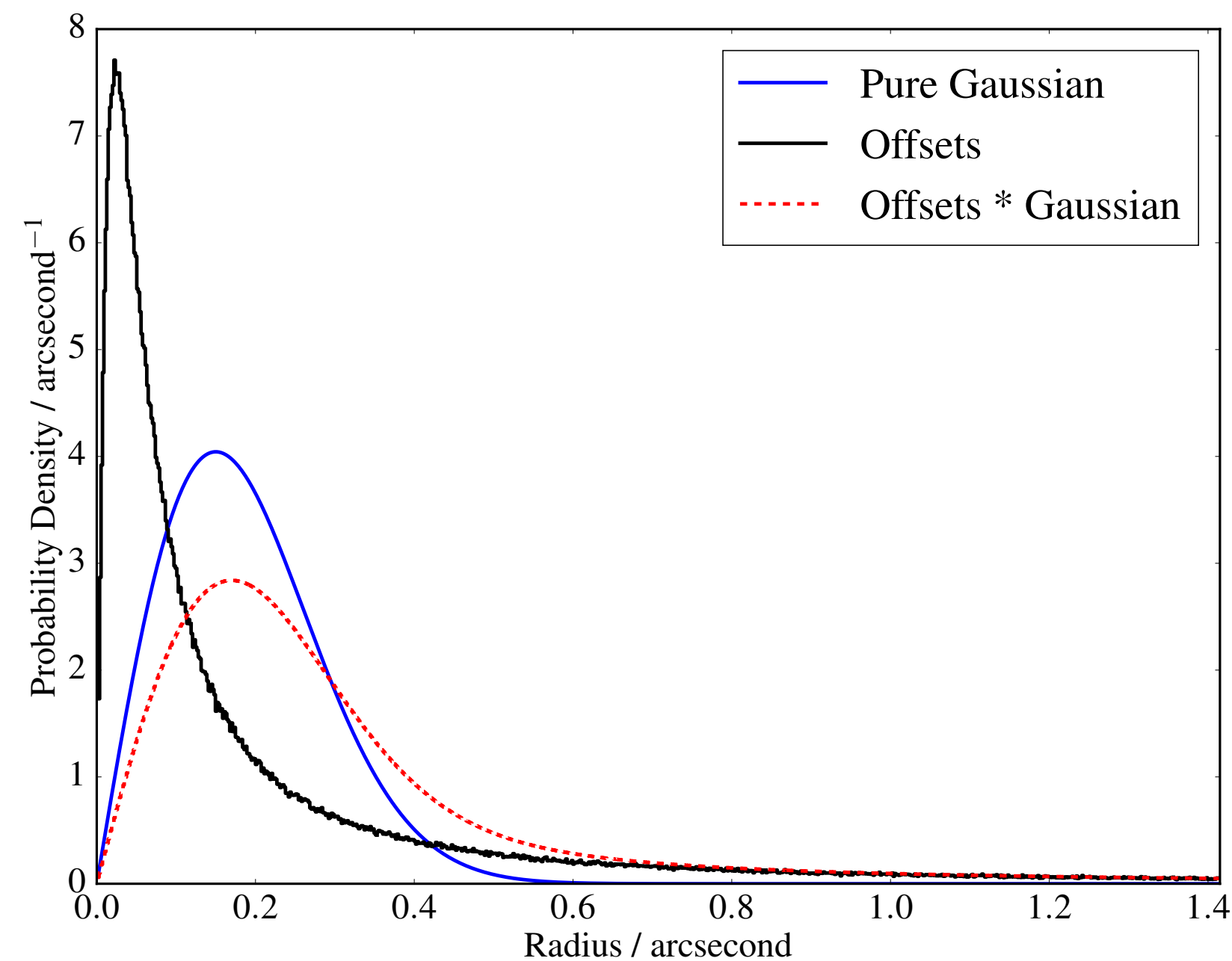
Matching Across Catalogues using the Astrometric Uncertainty Function and Flux

macauff is a cross-match algorithm that:

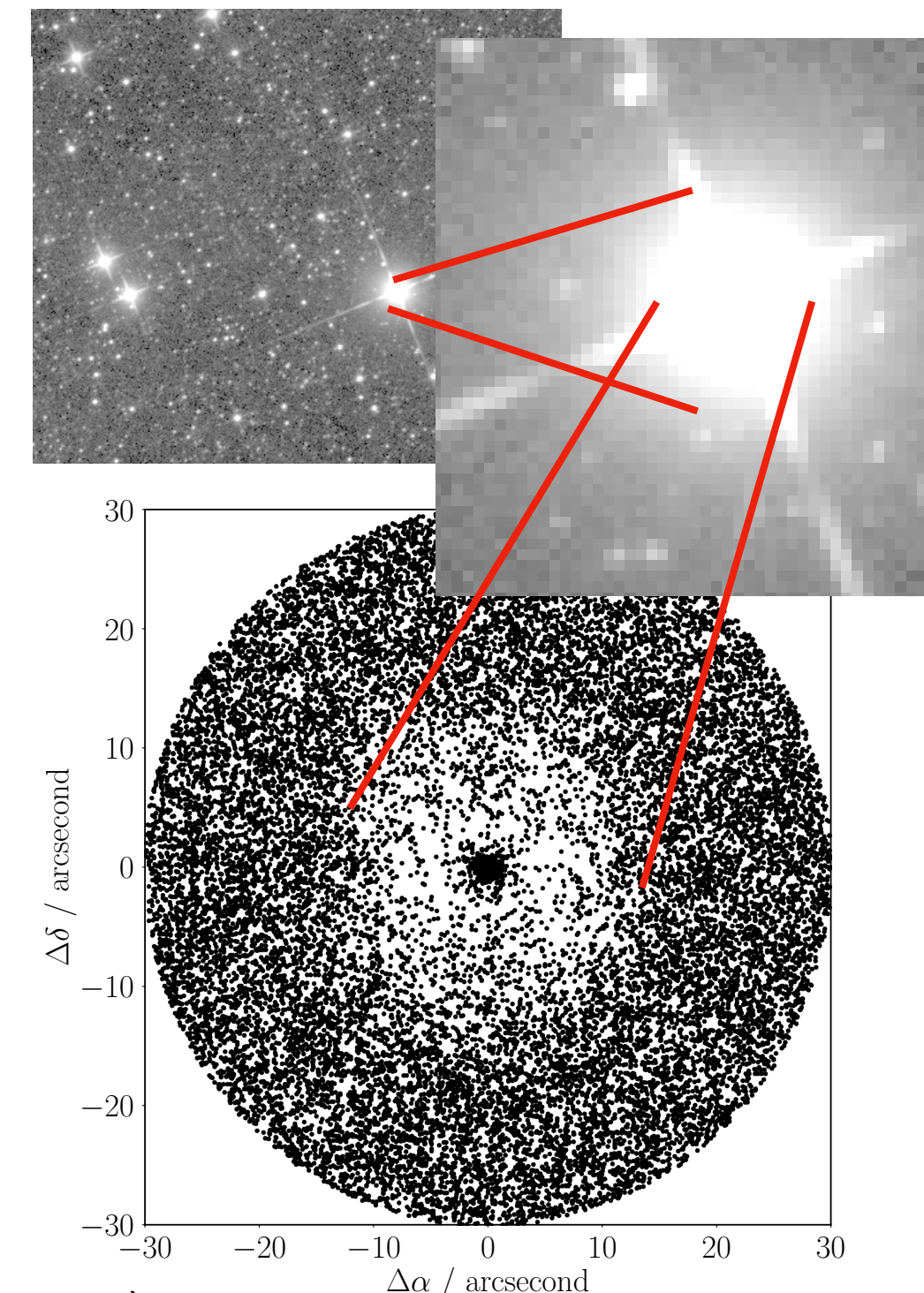
- is a “many-to-many” algorithm, solving for all potential competing counterparts simultaneously
- is fully Bayesian, providing relative strength of match pairings for the user
- includes the astrometric uncertainties of all objects in considering matches on positional grounds
- uses the measured colours of potential matches as a further discriminator of true and false matches
- **extends the “Astrometric Uncertainty Function”** to include more than just the *centroid* uncertainty, adding e.g.
 - unknown or weakly-known proper motion of stars
 - the **“Perturbation Component” of the AUF** — the effect of unresolved contaminants on measured positions



WISE - Wright et al. (2010)
Gaia DR2 - Gaia Collaboration, Brown A. G. A., et al. (2018)



Wilson & Naylor (2018b)



Wilson & Naylor (2017) Tom J Wilson @onoddil

Why Use Our Cross-Matches?

- 0) Getting cross-matches, even for “well behaved” fields**
- 1) Recovering additional sources missed by other match services — either in crowded fields (we recover up to twice as many *Gaia-WISE* matches than the *Gaia* best neighbour matches), or with our extension to unknown proper motion modelling as an extra systematic**
- 2) Finding “odd” objects, either using the inclusion vs non-inclusion of the photometry in the two match runs, or via the likelihood ratio space — separately-planned “real time” matching service for transient objects**
- 3) Removing e.g. IR excess or correcting for extinction-like crowding flux brightening, through Average Contamination; crucial for “1% photometry” in both precision *and* accuracy**
- 4) Corrections to inconsistent astrometric covariances, key for DP2**

How To Use Our Cross-Matches

(Or, how this impacts you on a day-to-day basis)

0) Getting cross-matches



Three tables per cross-match: merged catalogue dataset, and 2x non-match dataset (one per catalogue)

The screenshot shows a web application interface for astronomical data queries. At the top, there's a navigation bar with tabs like "Results", "DP1 & DP2 Images", "DP1 & DP2 Catalogs", "Alert Viewer", "Upload", and "Job Monitor". Below this, a "Tables" section shows a selected table collection "macaulf_gaiadr3xcatwise2020" with a description: "Match and non-match table for macaulf cross-matches of Gaia". To the right, a "Tables:" section lists available tables, including "macaulf_gaiadr3xcatwise2020.gaiadr3nonmatches_astro_photo". The main area is titled "Enter Constraints" and has a "Spatial" tab selected. It includes fields for "Shape Type" (Cone Shape or Polygon Shape), "Coordinates or Object Name" (with a "Try NED then Simbad" button), "Radius" (set to 10 arcseconds), and "Position Column" (set to "Unspecified Column(s)"). Below these are fields for "Lon Column" and "Lat Column". On the right, an "Output Column Selection and Constraints" table is visible, listing various columns like "bp", "chunkid", "g", "gaiadr3_rp", etc., with their respective data types. At the bottom, there are buttons for "Search", "Row Limit" (set to 50000), "Title" (set to "macaulf_gaiadr3xcatw..."), and "Populate and edit ADQL".

Example columns:

- Designations of the two sources (e.g., WISE J... and *Gaia* EDR3...)
- RA and Dec (or Galactic l/b) of the two sources
- Magnitudes (corrected for necessary effects, such as e.g. *Gaia*) in all bandpasses for both objects
- **Match probability** — probability of the most likely permutation (see equation 26 of Wilson & Naylor 2018a)
- **Eta** - Photometric likelihood ratio (counterpart vs non-match probability, just for brightnesses; see eq37 of WN18a)
- **Xi** - Astrometric likelihood ratio (just position match/non-match comparison; see eq38 of WN18a)
- **Average contamination** - simulated mean (percentile) brightening of the two sources, based on number density of catalogue
- **Probability of sources having blended contaminant** above e.g. 1% relative flux

We will provide two match runs per catalogue pair match: one with, and one without, the photometry considered, to allow for the recovery of sources with “weird” colours but otherwise agreeable astrometry

The Rubin “Super-Match”

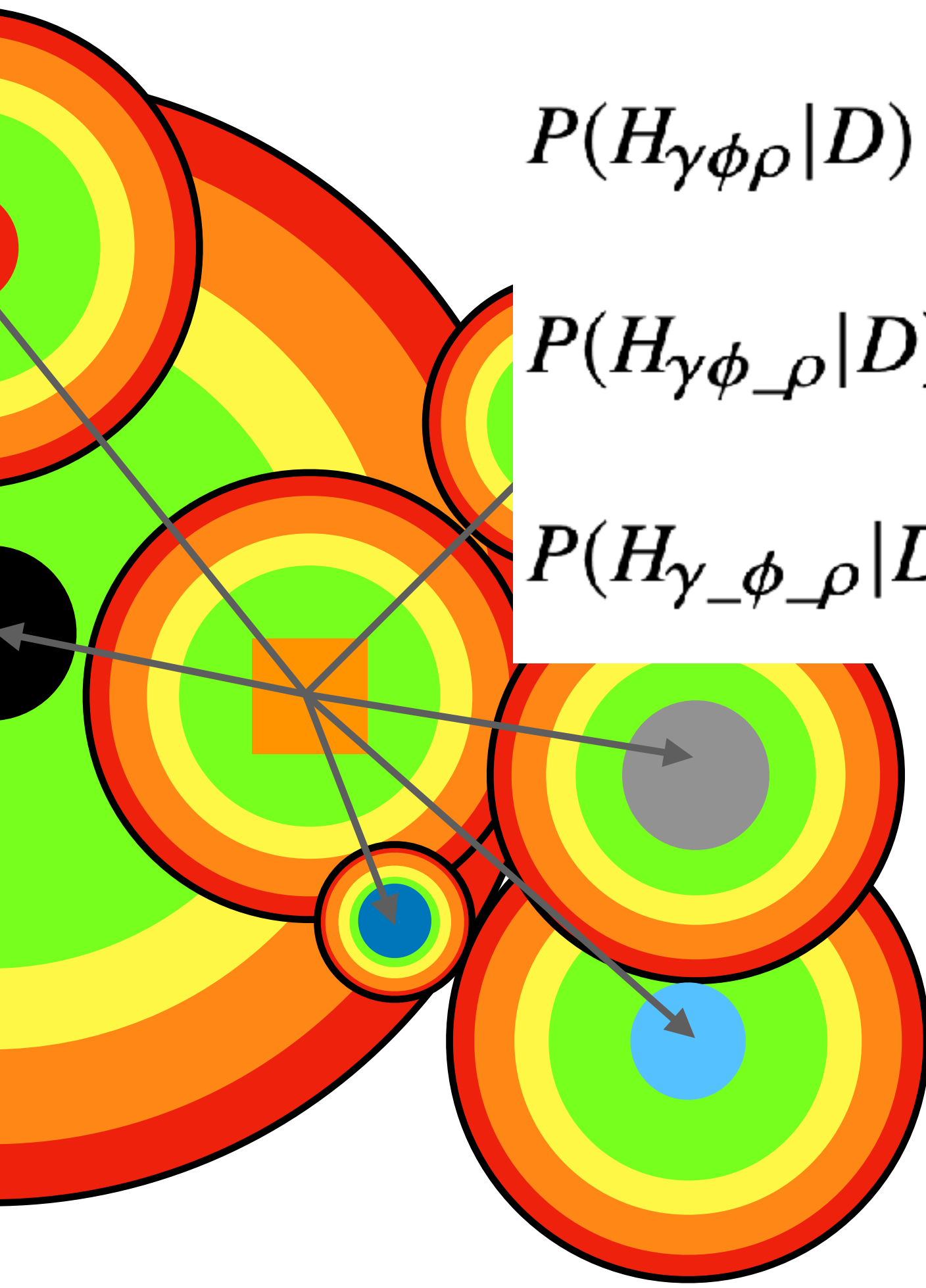
Bringing Independent Results together to Notify of Associations across Multiple catalogues

0) Getting cross-matches

LSST -> *Gaia*, *WISE*, *VISTA*, *Euclid*, *SDSS*, ... matches

Quick and easy construction of spectral energy distributions for each LSST source

Includes SED probabilities, individual match reliability, contamination statistics etc.



$$P(H_{\gamma\phi\rho}|D) = \frac{1}{K} N_{\gamma\phi} G_{\gamma\phi} N_{\gamma\rho} G_{\gamma\rho} = A_{\gamma\phi} A_{\gamma\rho},$$

$$P(H_{\gamma\phi_ \rho}|D) = \frac{1}{K} N_{\gamma\phi} G_{\gamma\phi} N_{\gamma} N_{\rho} = B_{\gamma\phi} A_{\gamma\rho},$$

$$P(H_{\gamma_ \phi_ \rho}|D) = \frac{1}{K} N_{\gamma} N_{\phi} N_{\gamma} N_{\rho} = B_{\gamma\phi} B_{\gamma\rho},$$

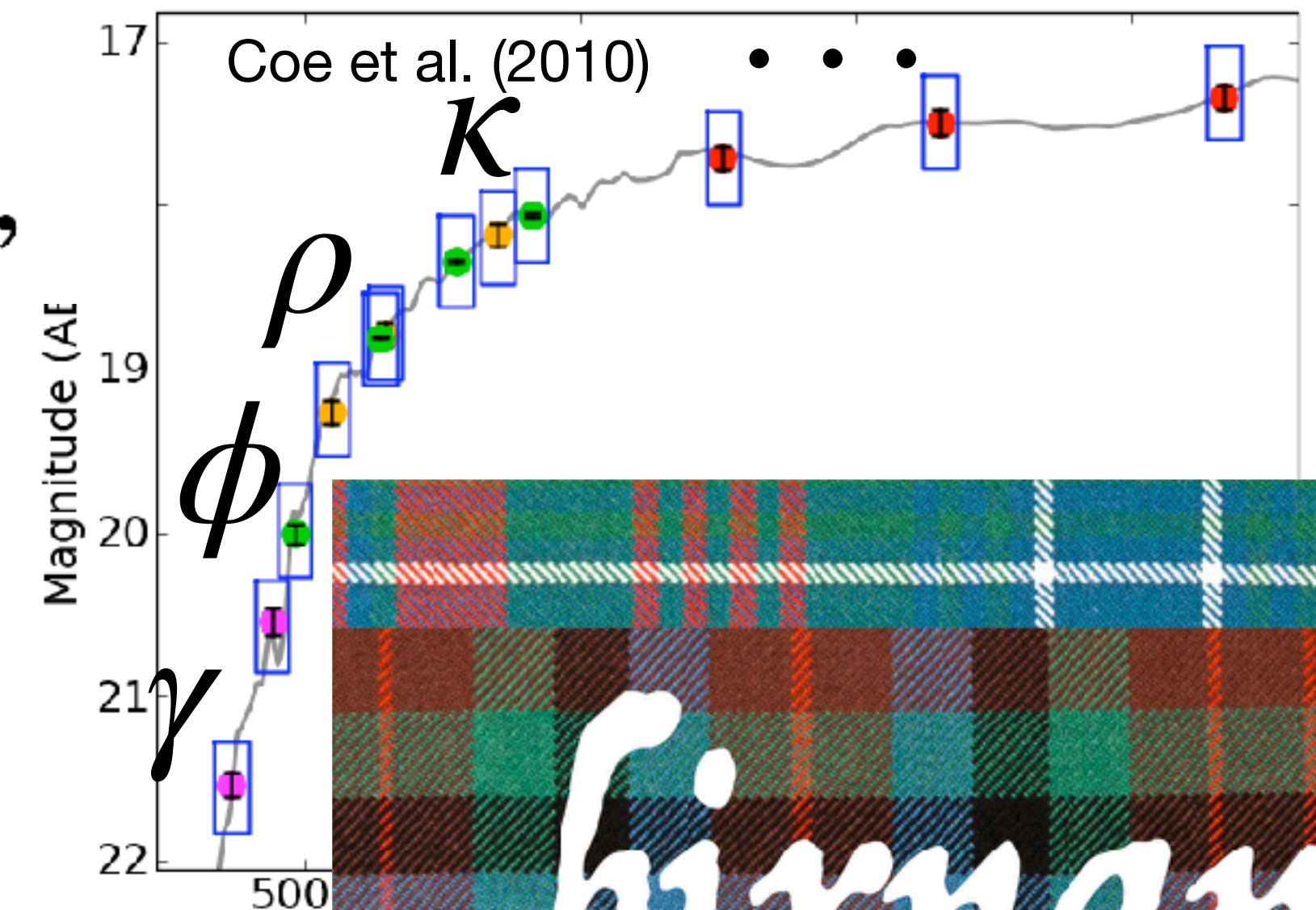
$$A_{\gamma\phi} = \frac{N_{\gamma\phi} G_{\gamma\phi}}{N_{\gamma} N_{\phi} + N_{\gamma\phi} G_{\gamma\phi}}$$

$$B_{\gamma\phi} = \frac{N_{\gamma} N_{\phi}}{N_{\gamma} N_{\phi} + N_{\gamma\phi} G_{\gamma\phi}}$$

Wilson & Naylor (2018a)

Wilson & Naylor (in prep.)

Pineau et al. (2017)

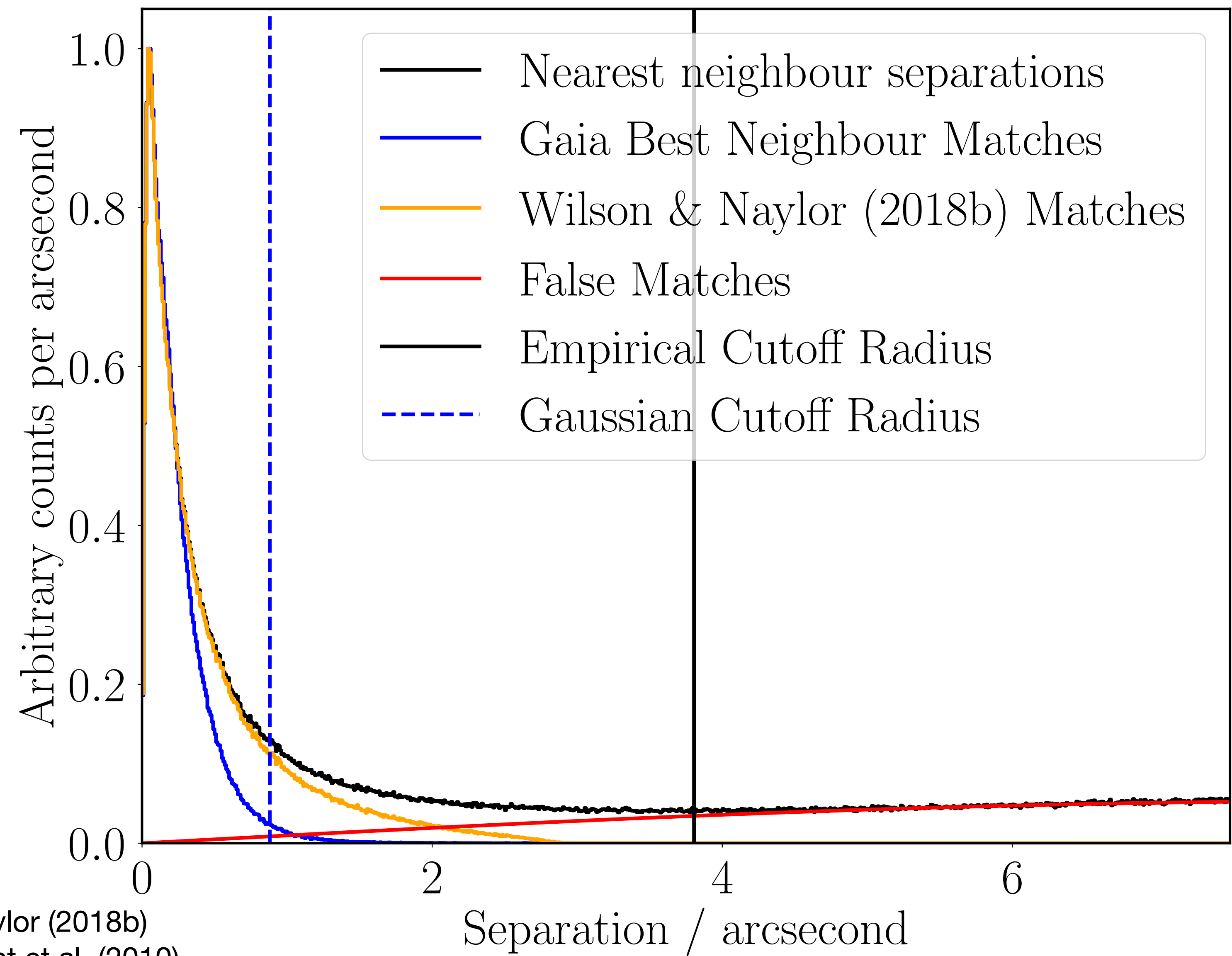


<https://github.com/macauuff/macauuff>

<https://github.com/macauuff/birnam>

Tom J Wilson @onoddil

Finding Matches Others Won't



1) Recovering additional sources

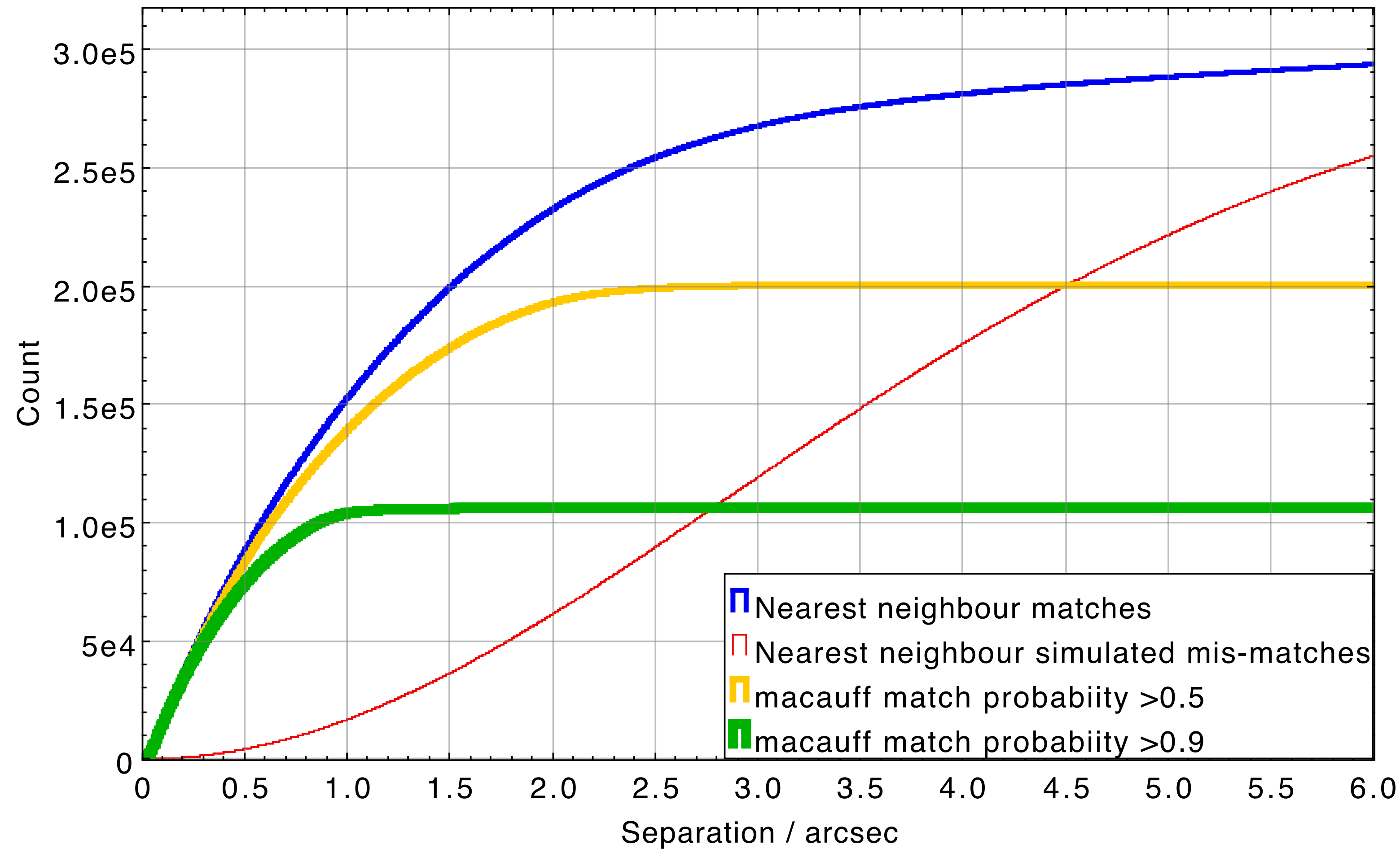
If perturbation due to unresolved contaminants was not taken into account, we would be incorrectly led to believe 50% of *Gaia-WISE** sources were not matches!

*“*Euclid*-Rubin”

Wilson & Naylor (2018b)
WISE - Wright et al. (2010)
Gaia matches - Marrese et al. (2019)
Gaia DR2 - Gaia Collaboration, Brown A. G. A., et al. (2018)

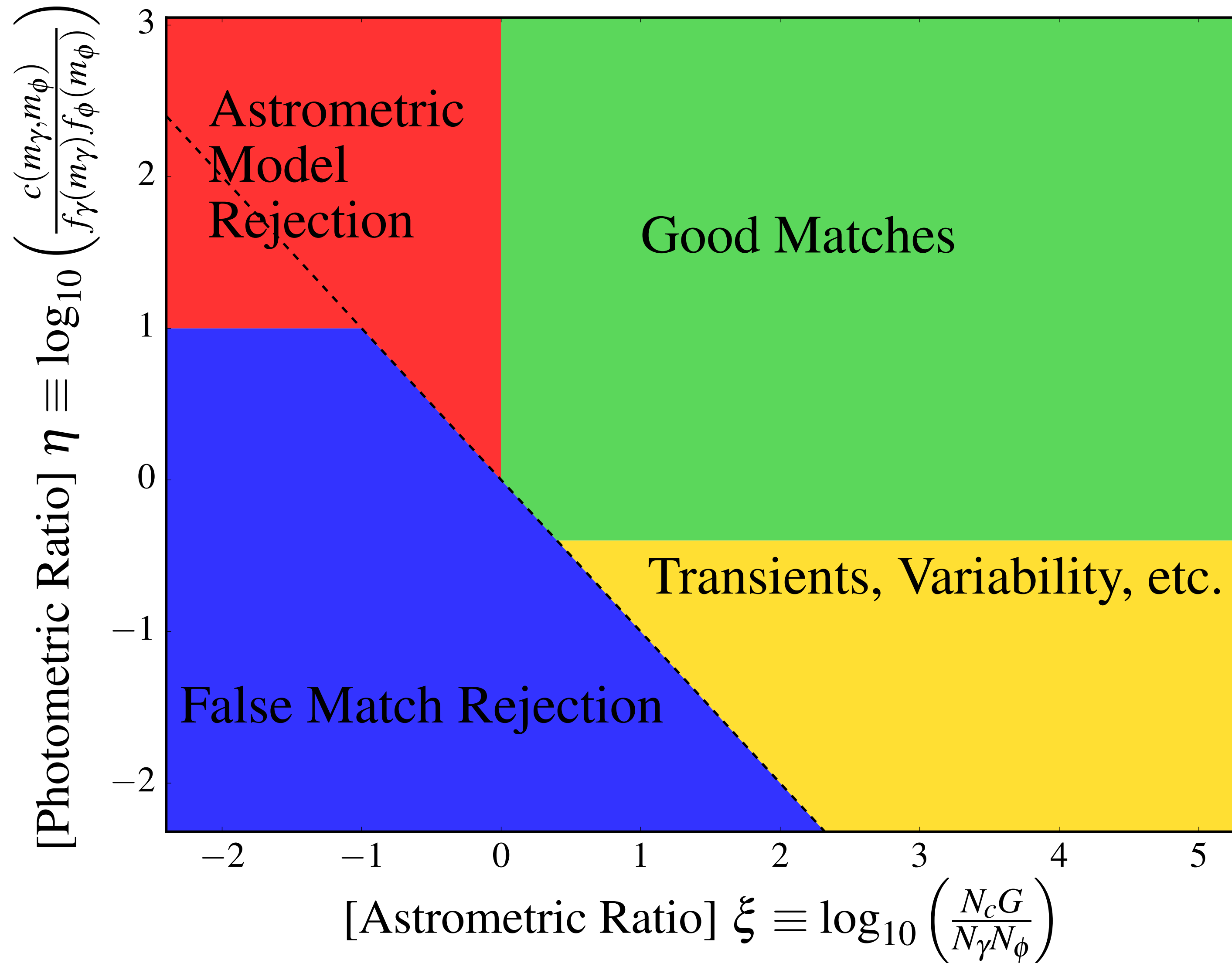
Robust Datasets with Match Probability

1) Recovering additional sources



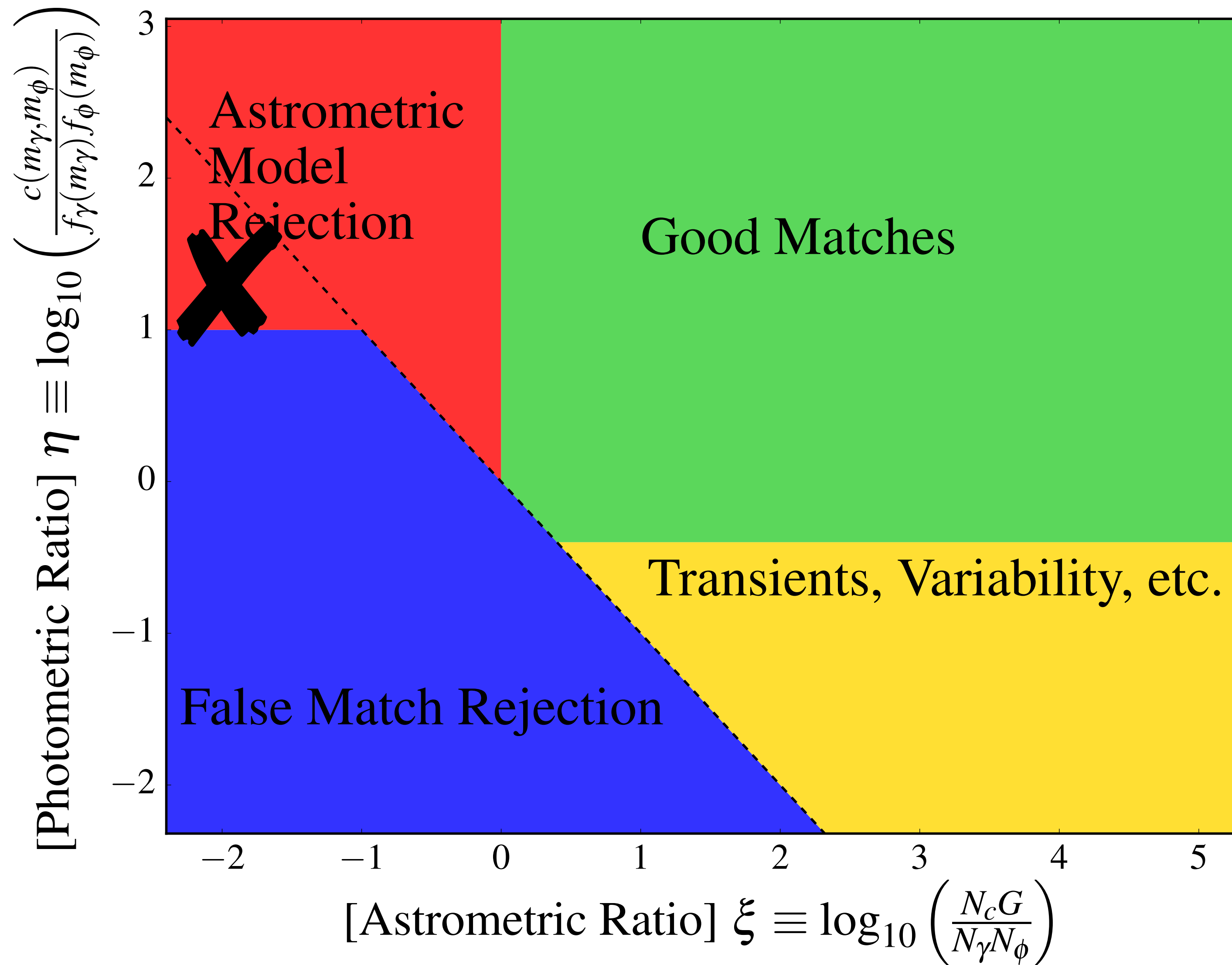
- Create a pure, if perhaps incomplete, sample, using e.g. `WHERE match_p > 0.999` to find only confident matches.
- Or, alternatively a lower threshold such as `match_p > 0.5` can be beneficial. Tests of a young-star classifier showed that including lower, inclusive match probabilities helped to remove older stars contaminating the data.

The Likelihood Ratio Phase Space



2) Finding “odd” objects

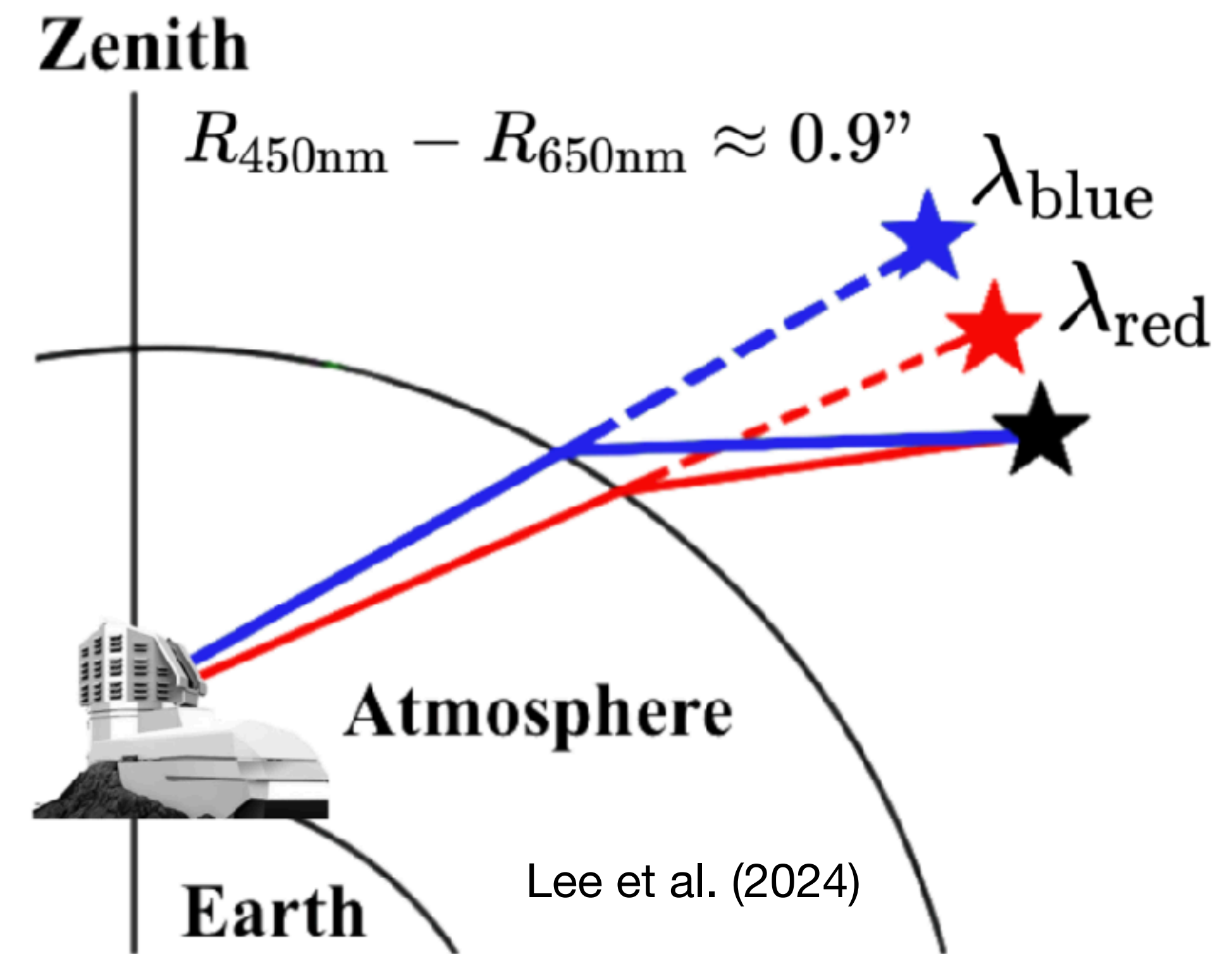
The Likelihood Ratio Phase Space



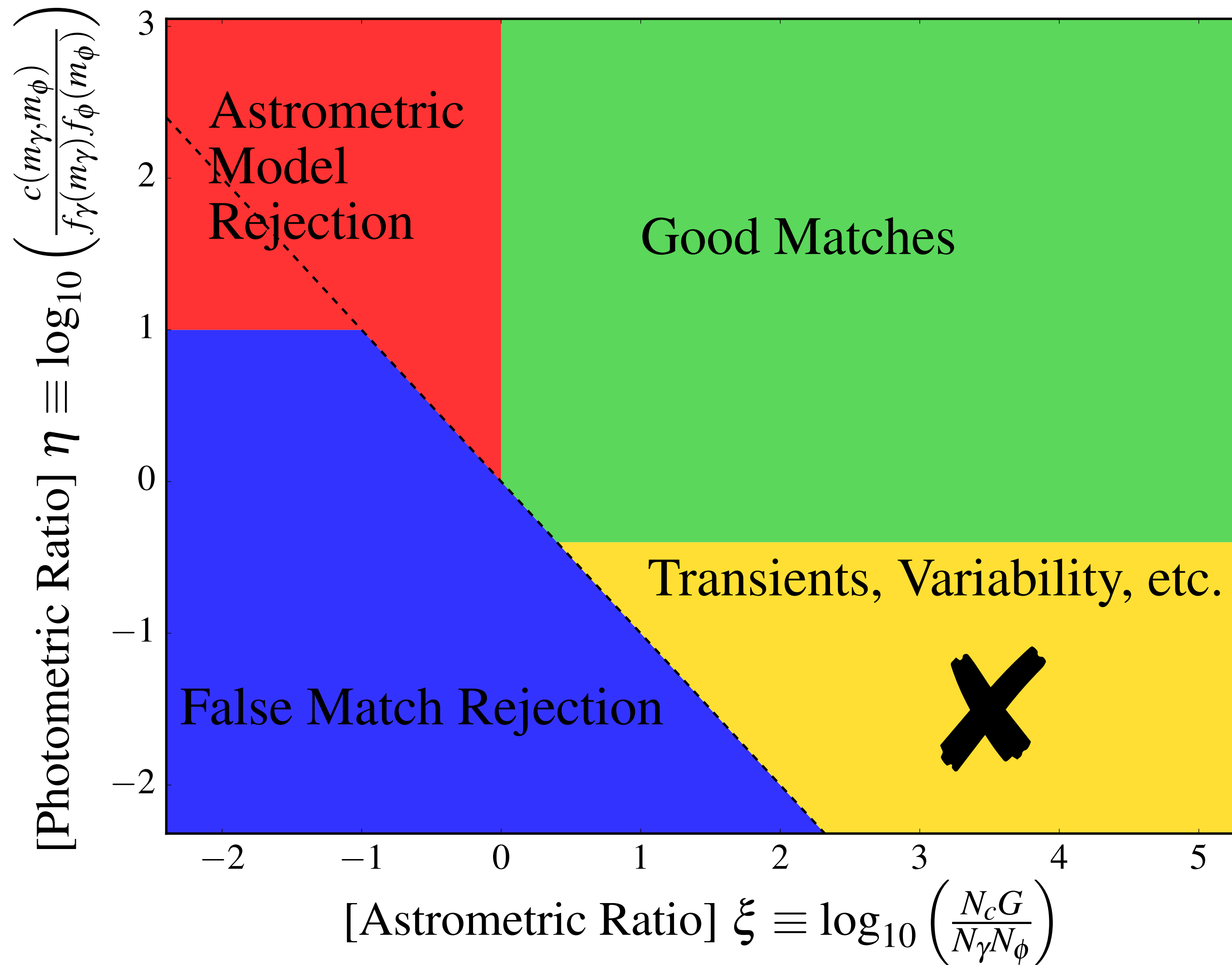
2) Finding “odd” objects

$$\Delta \mathbf{x}^w = K_b c \tan z \hat{\mathbf{p}}$$

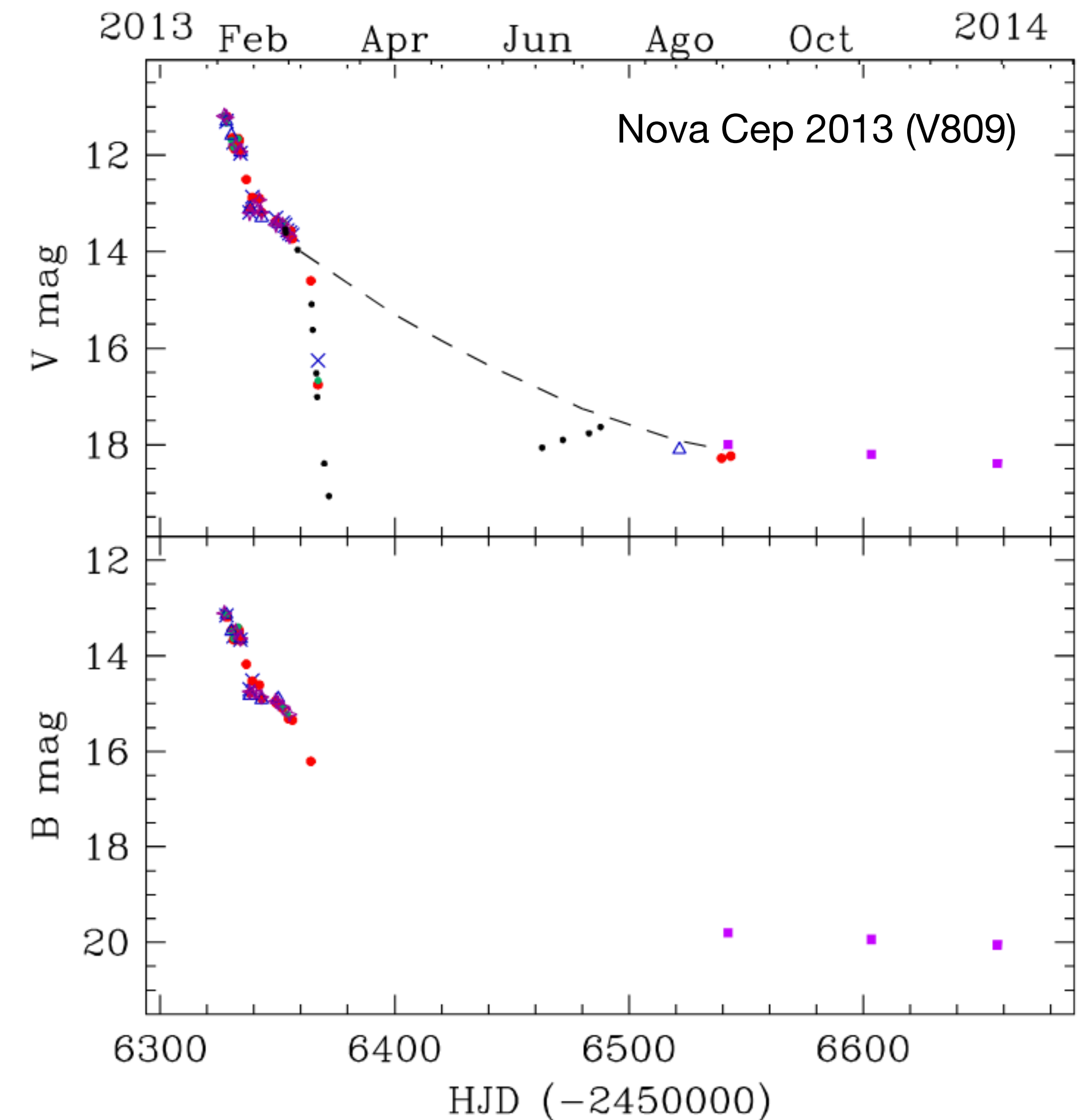
e.g. gbdes, Bernstein et al. (2017)



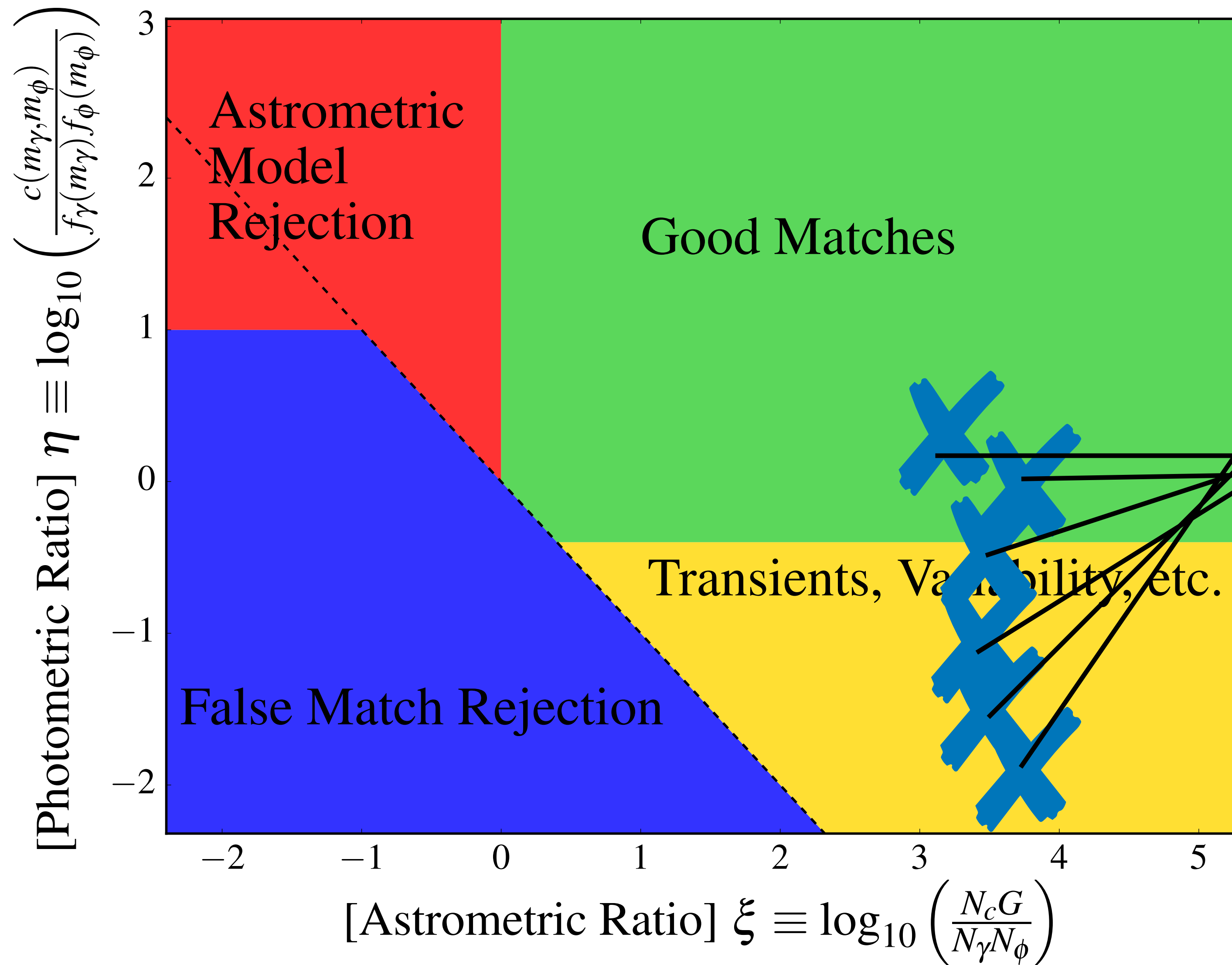
The Likelihood Ratio Phase Space



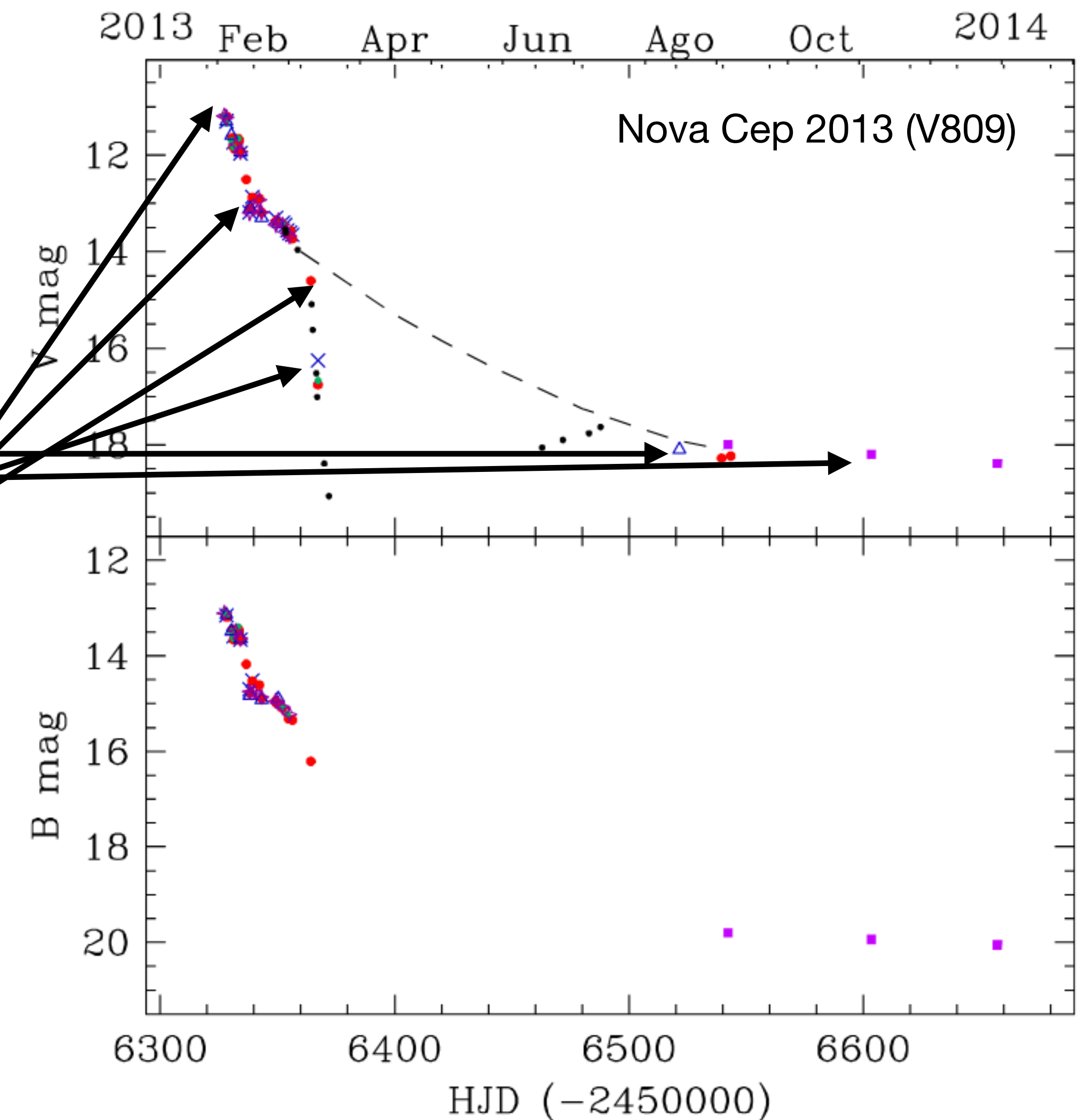
2) Finding “odd” objects



The Likelihood Ratio Phase Space



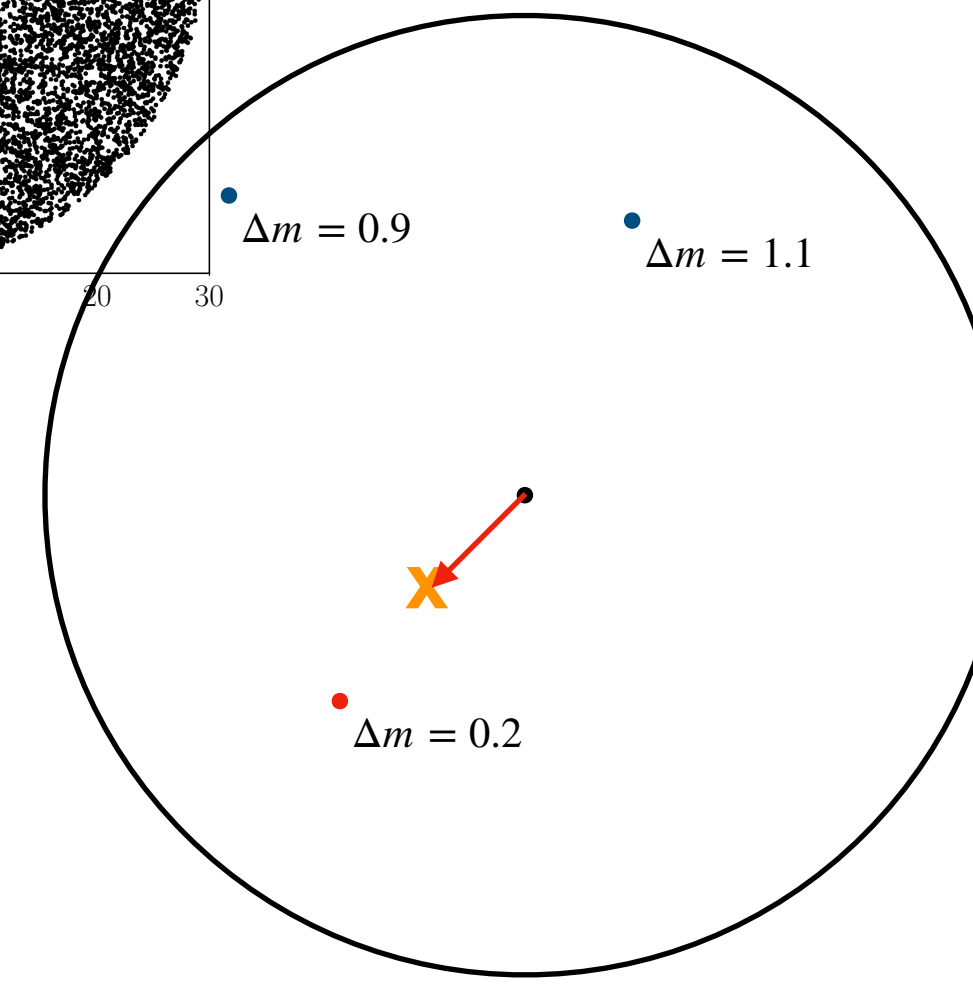
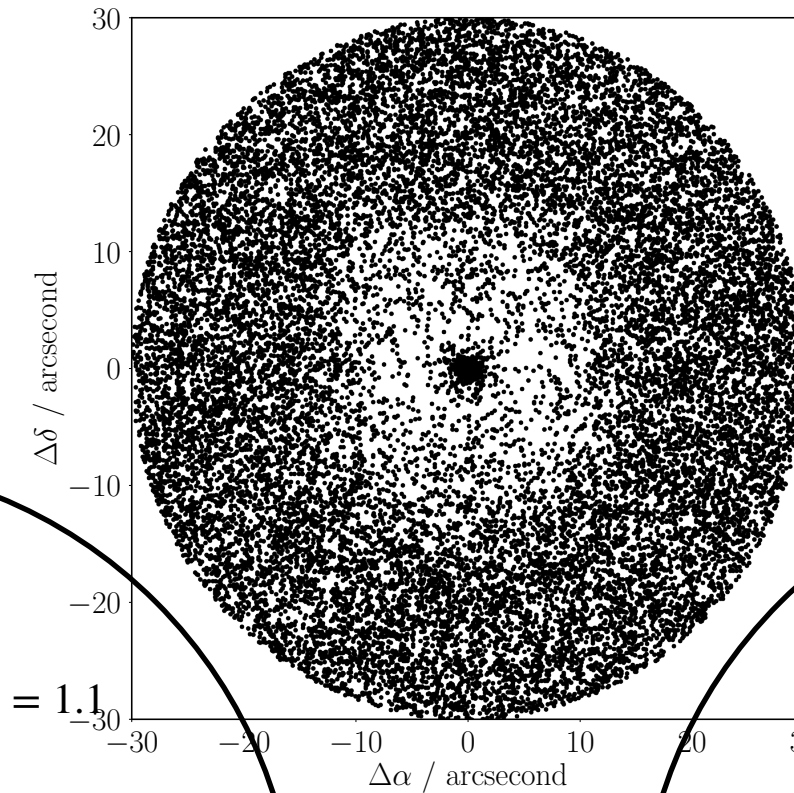
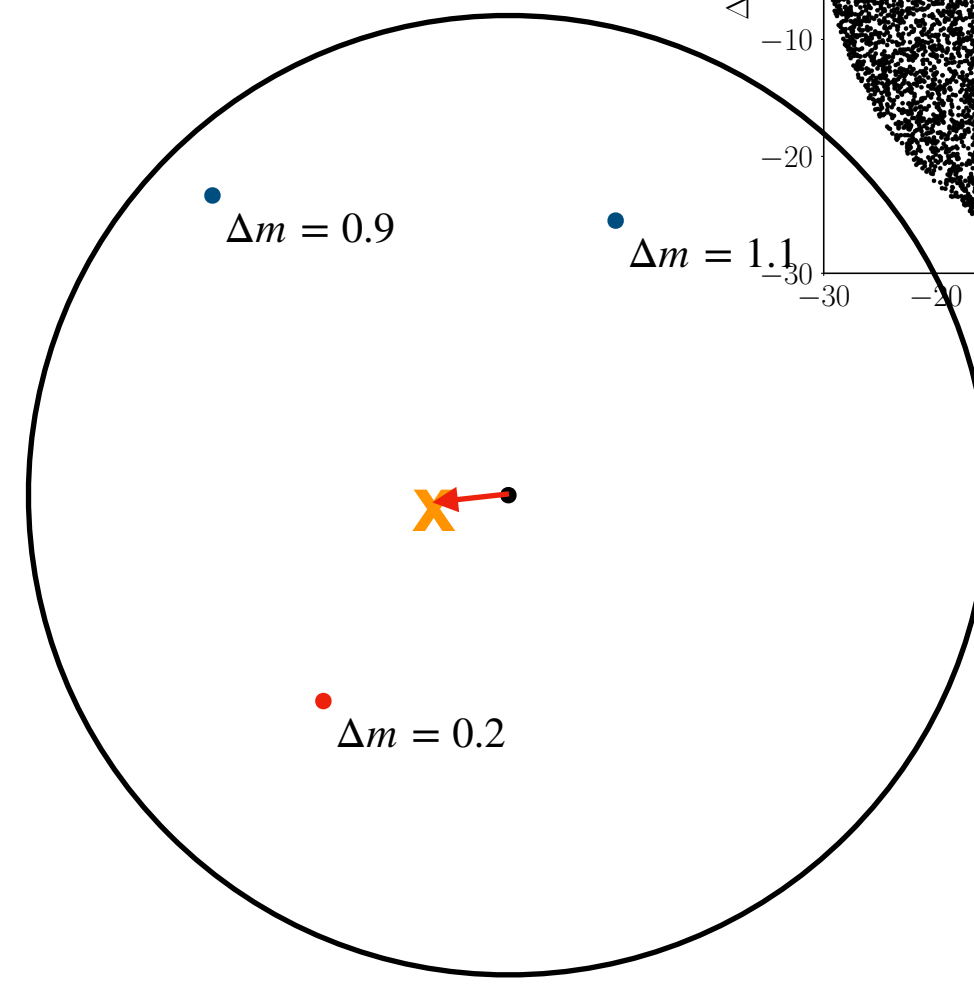
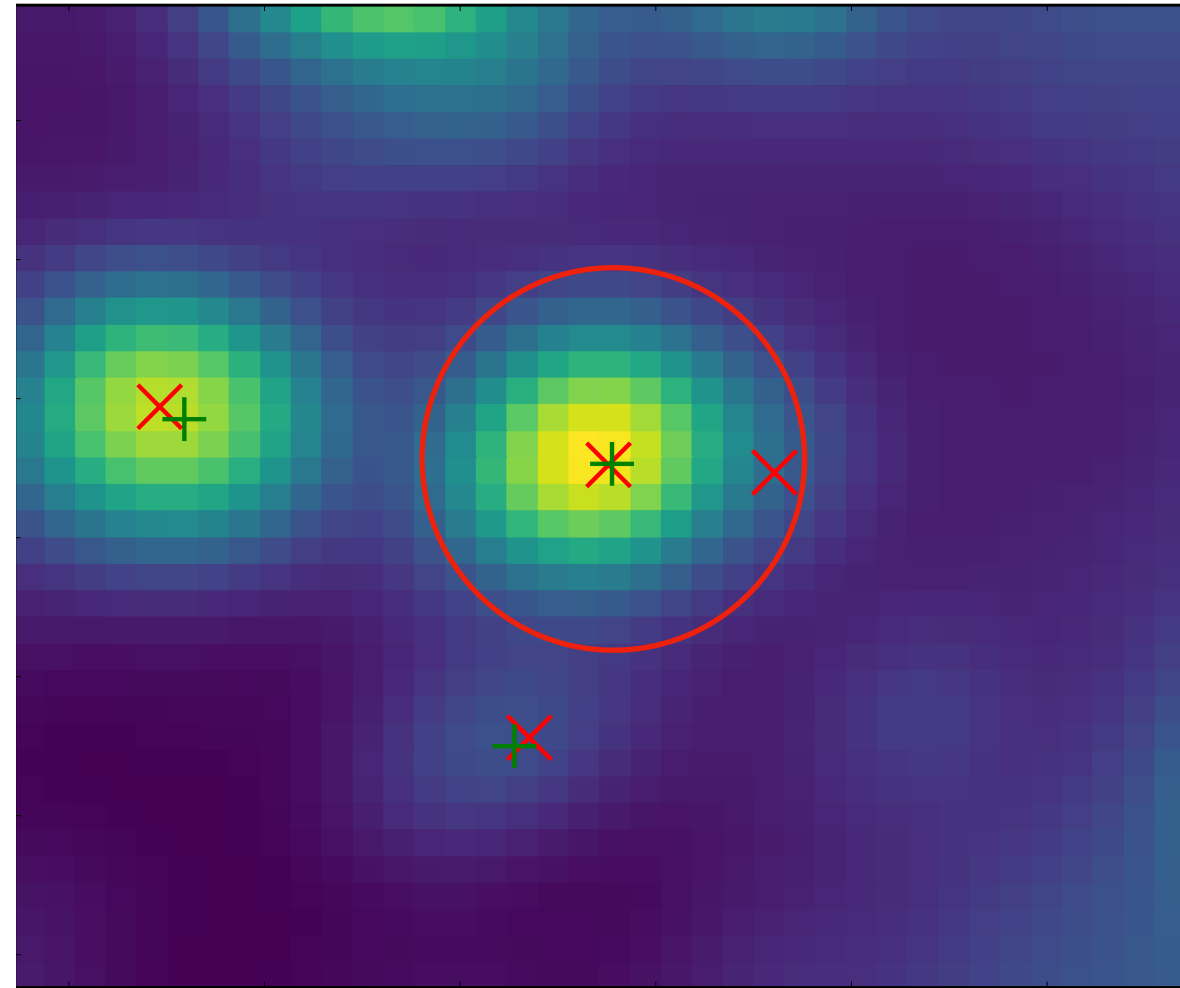
2) Finding “odd” objects



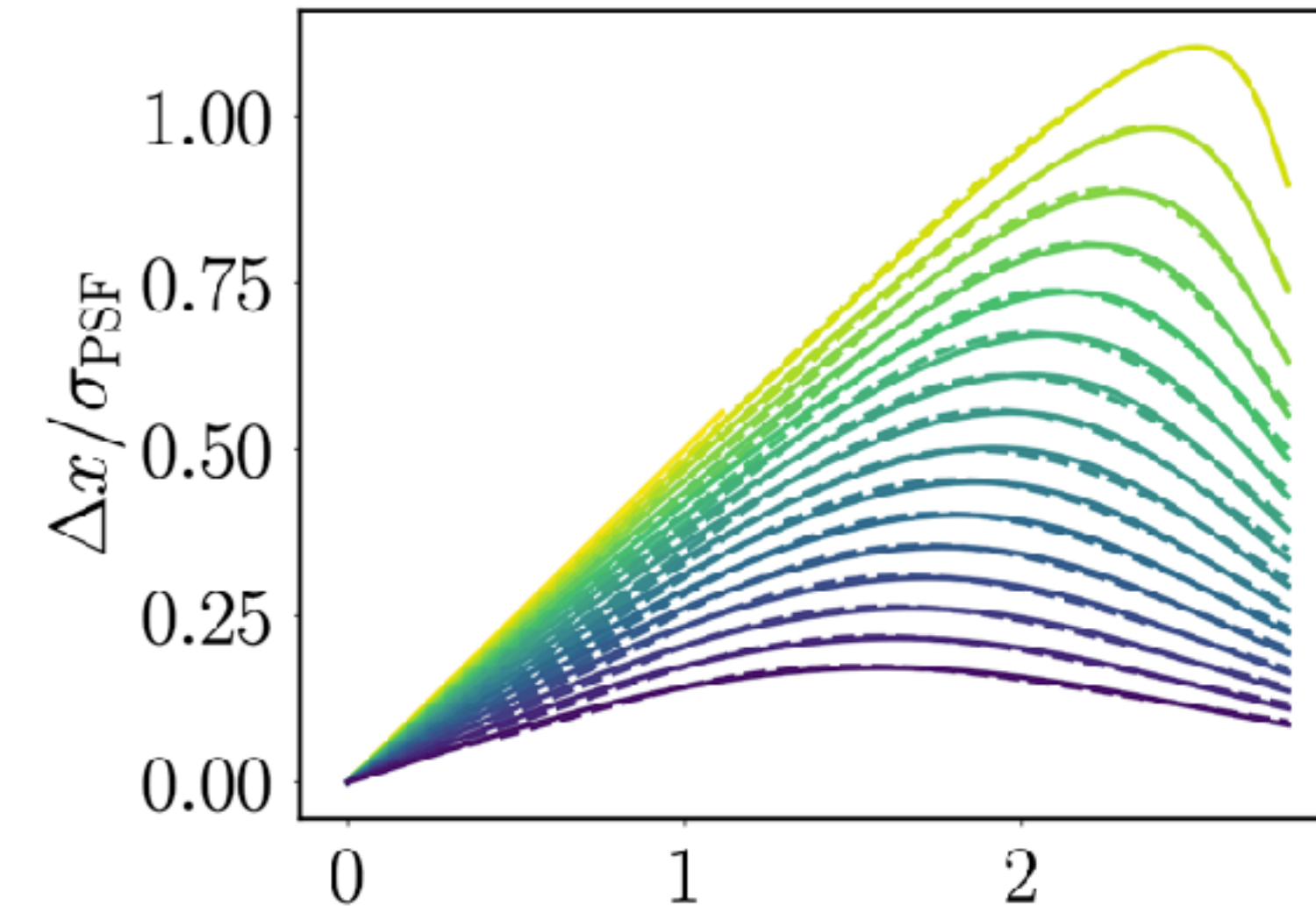
Crowded-Field Photometry

3) Correcting flux brightening

High SNR PSF or Aperture Photometry



Low SNR PSF Photometry



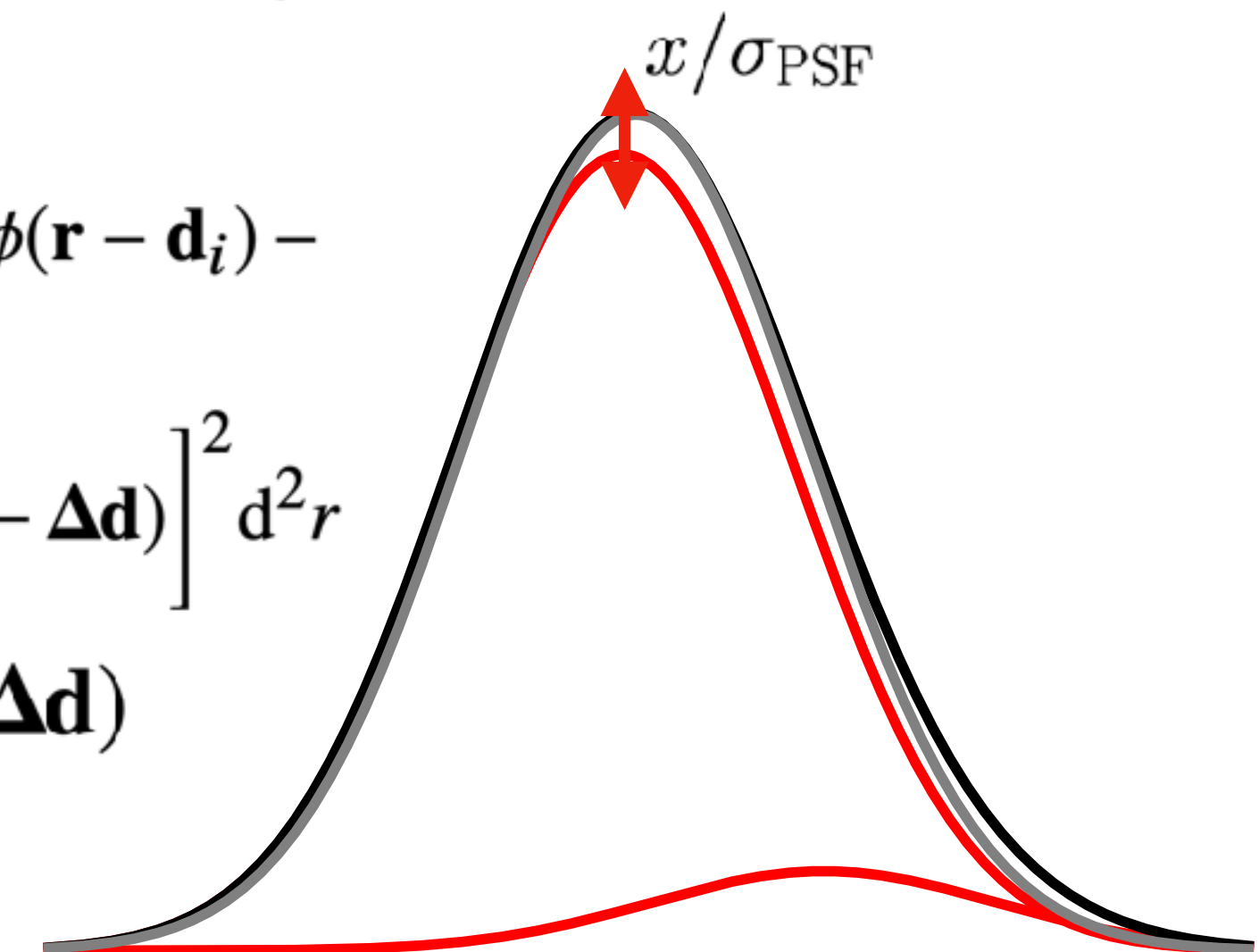
(This raises questions about the validity of quoting photometric statistical precisions if objects are systematically biased, and SED fitting in general in crowded fields)

$$\Delta x = \frac{\sum_i f_i x_i}{1 + \sum_i f_i}$$

$$\Delta f = \sum_i f_i$$

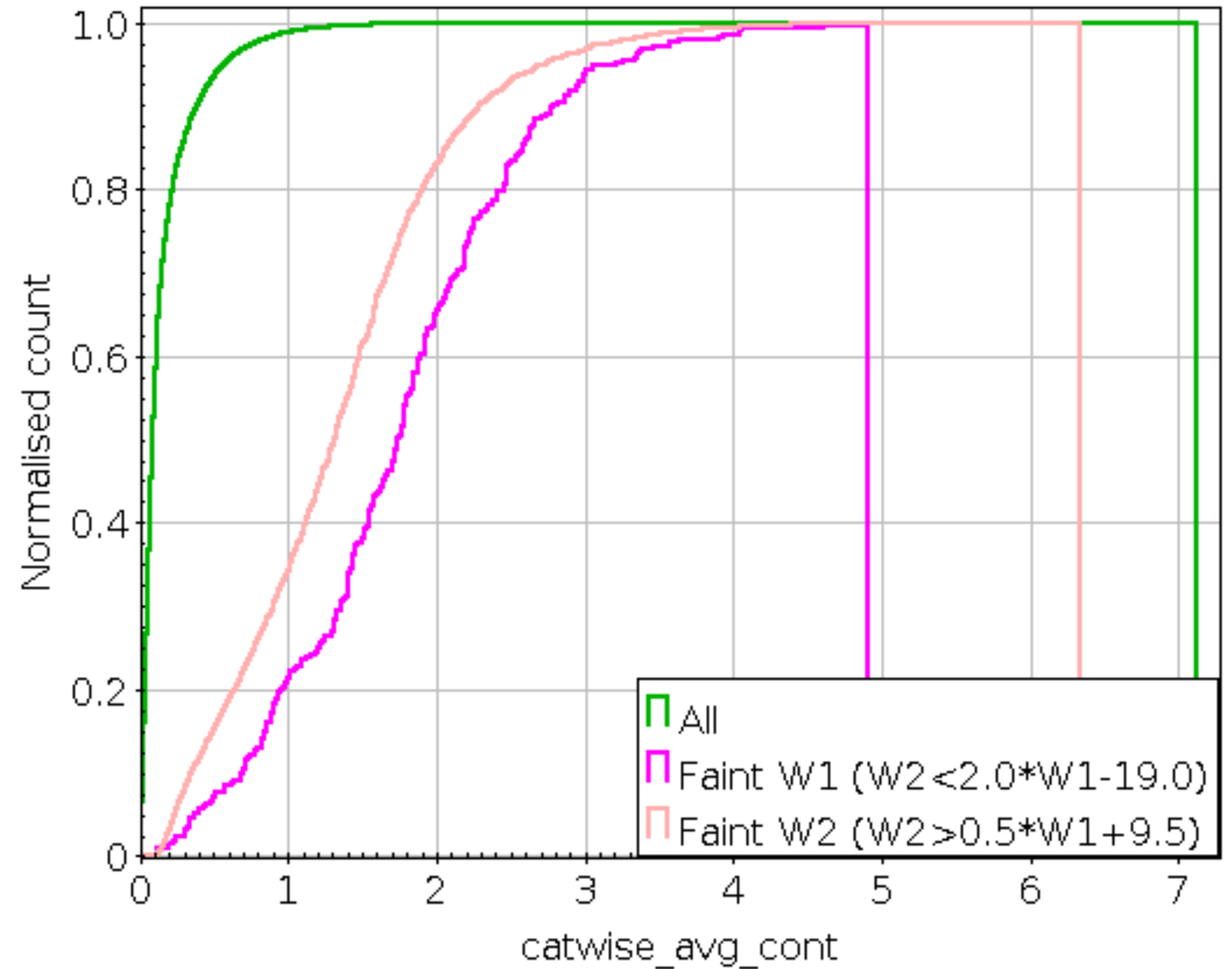
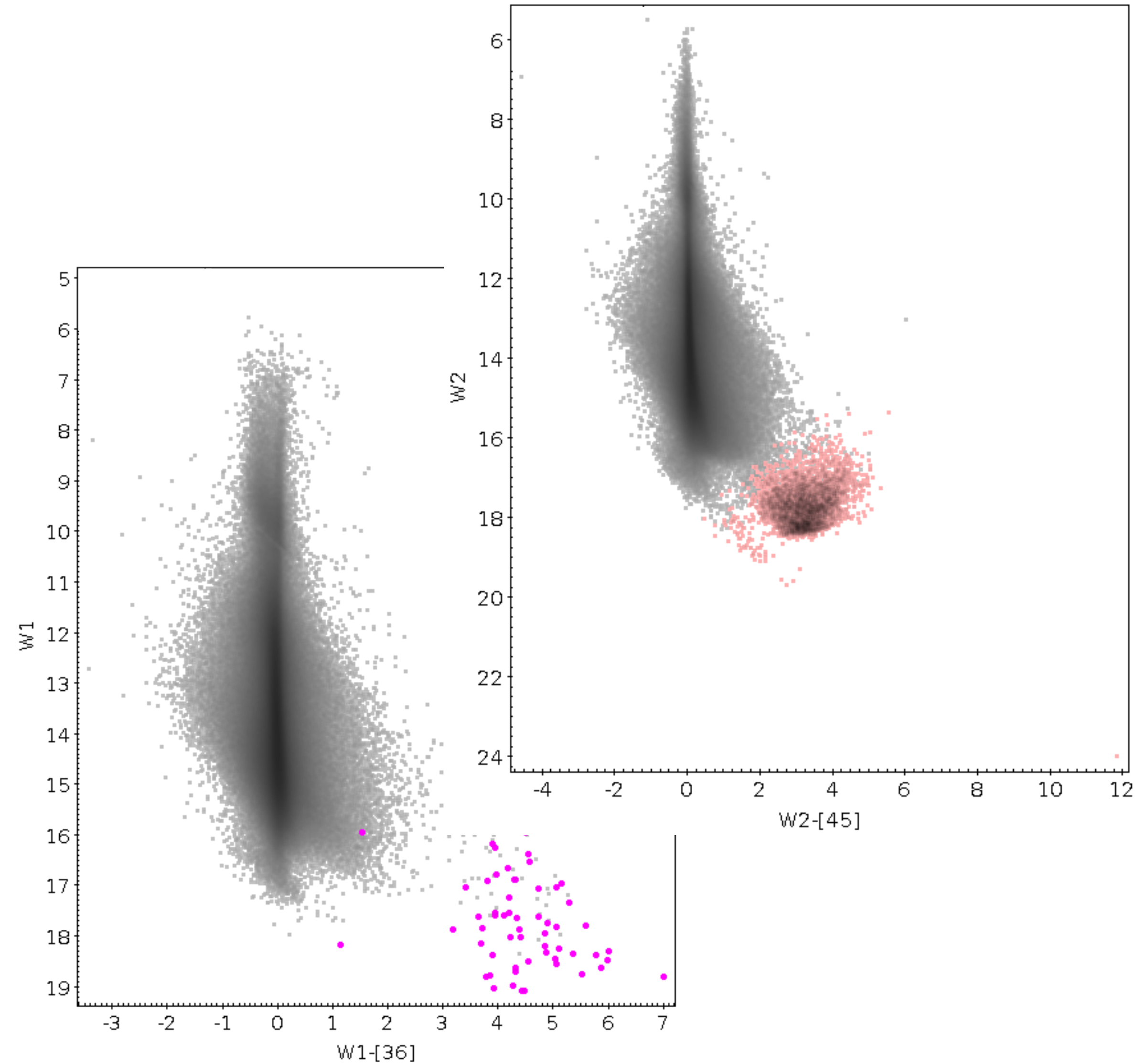
$$\log \mathcal{L} = -\frac{1}{2} \times L \int_{-\infty}^{\infty} \left[\phi(\mathbf{r}) + \sum_i f_i \phi(\mathbf{r} - \mathbf{d}_i) - (1 + \Delta f) \phi(\mathbf{r} - \Delta \mathbf{d}) \right]^2 d^2 r$$

$$\Delta f = \psi'(\Delta \mathbf{d}) - 1 + \sum_i f_i \psi'(\mathbf{d}_i - \Delta \mathbf{d})$$



Using Average Contamination

3) Correcting flux brightening



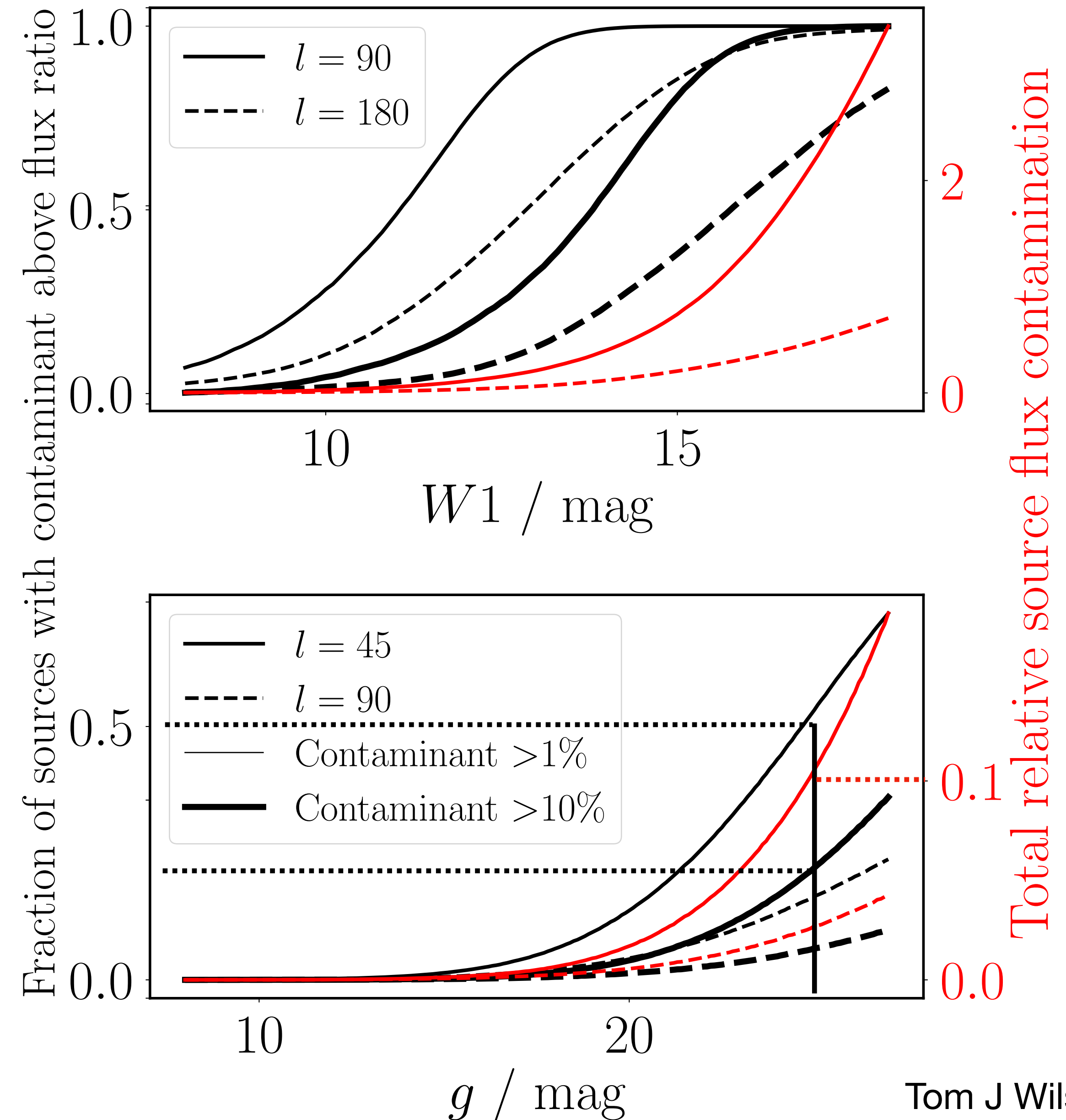
Using Average Contamination

3) Correcting flux brightening

Typical, single visit images in near-Bulge regions of the Plane will have:

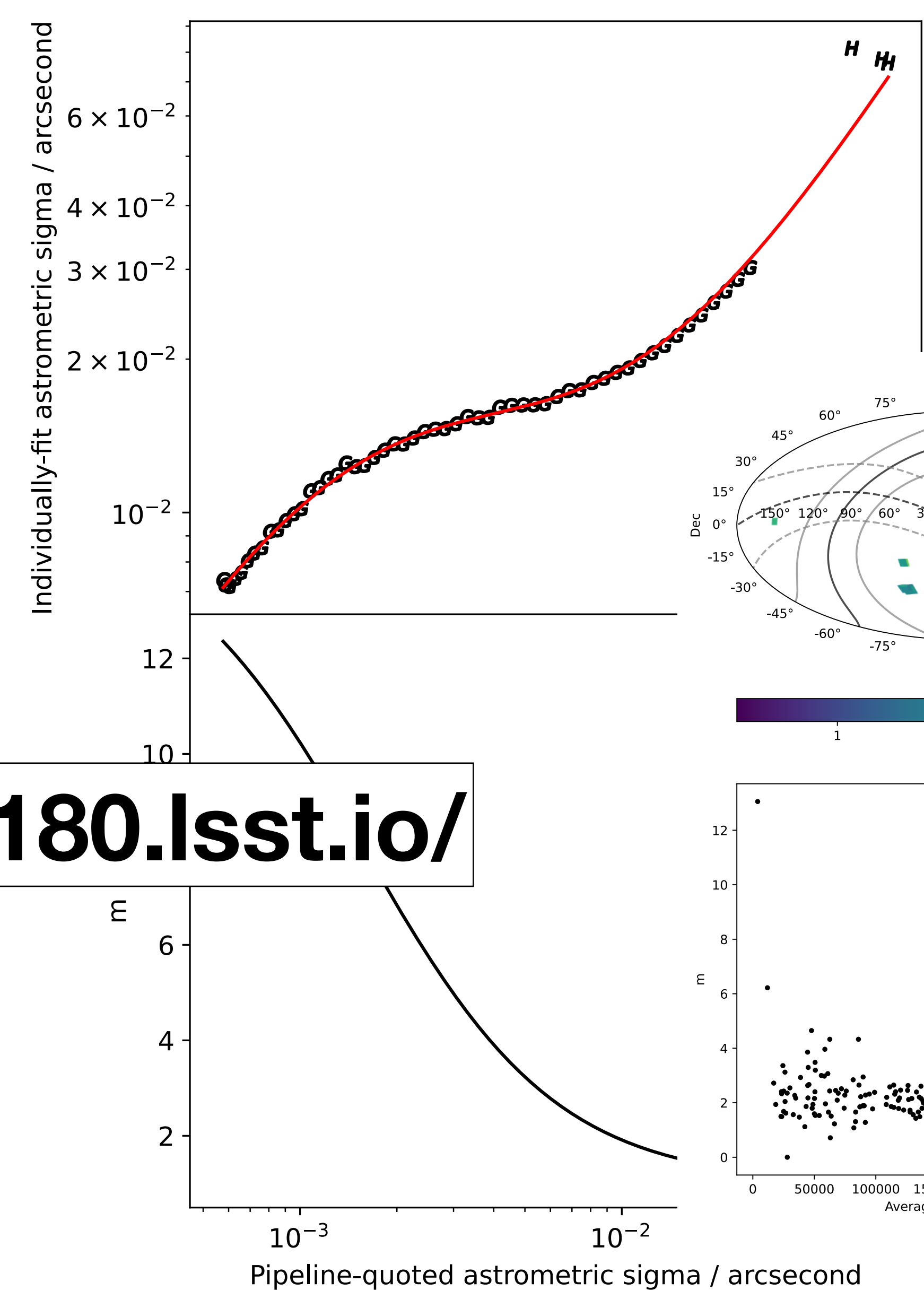
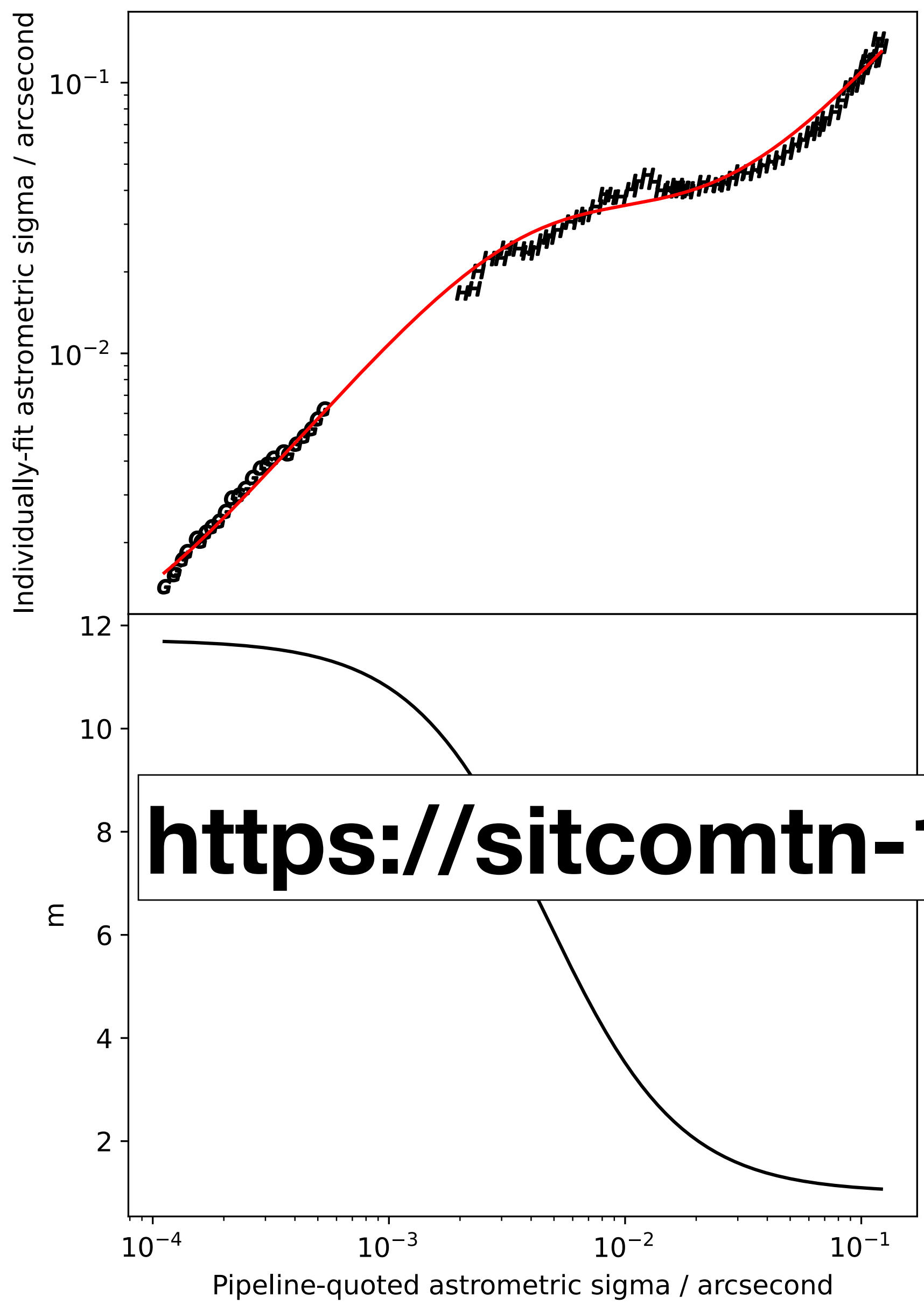
- **50% of objects with at least one >1% flux object in their PSF**
- **20% of objects with a >10% relative flux object contaminating them**
- **an average 10% total “extra” flux**

(the Bulge will be much more crowded! Nearest-neighbour matching won't work there, but neither will probabilistic matching without taking this effect into account...)

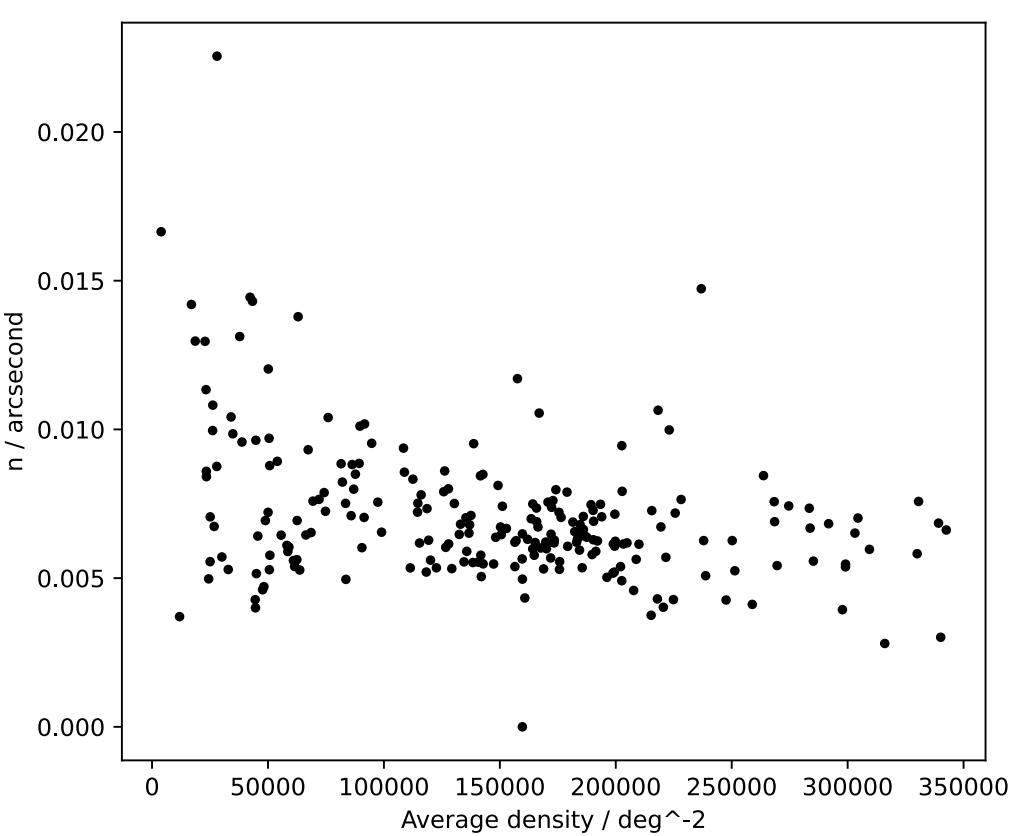
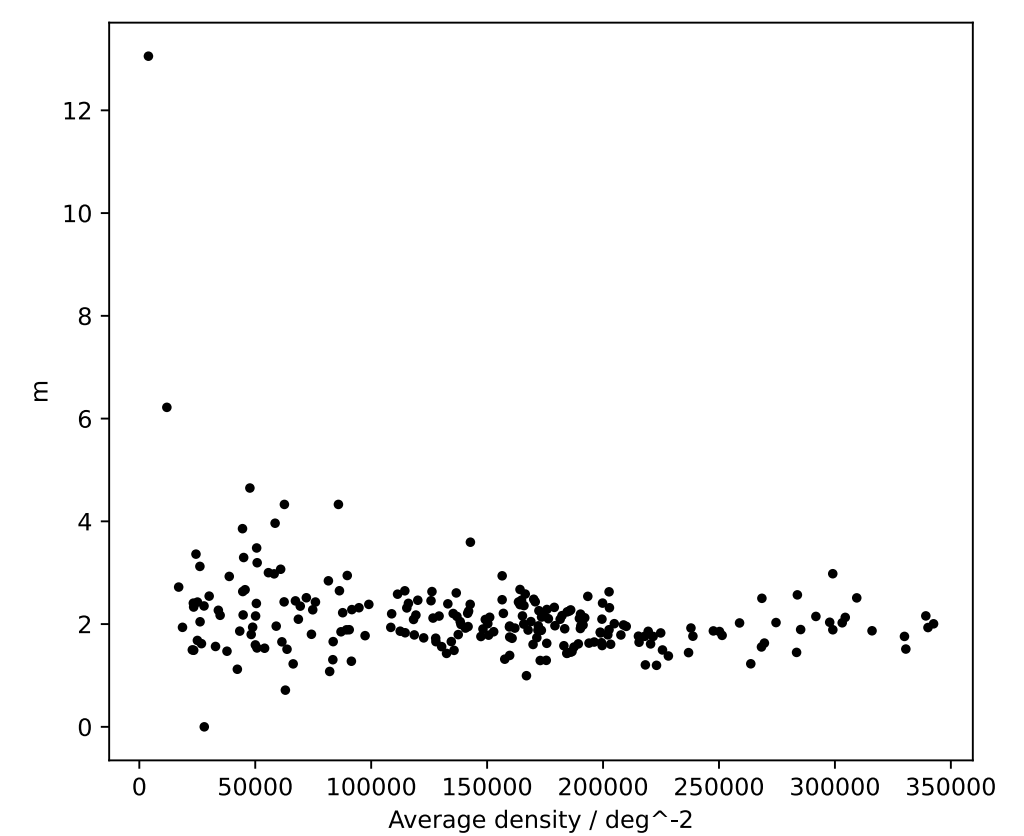
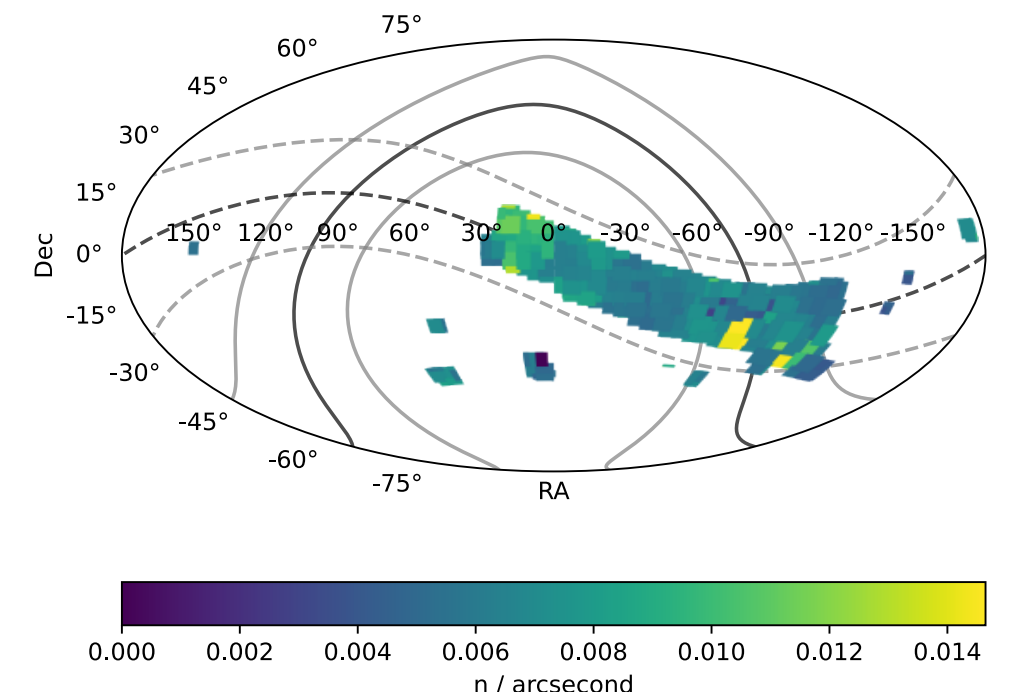
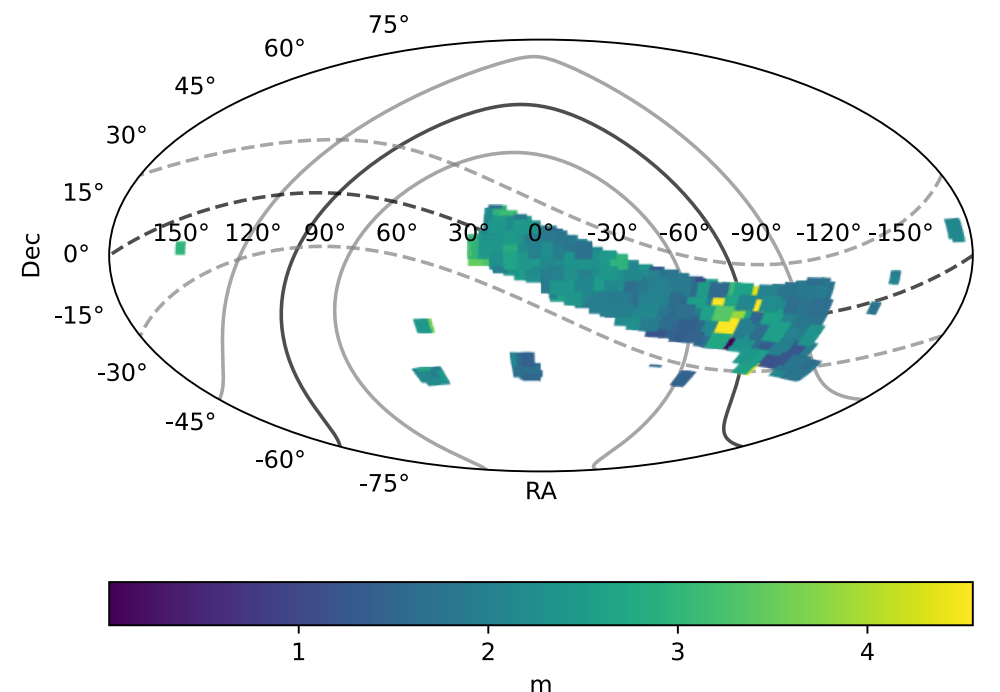


Fixing Catalogue Astrometry

4) Corrections to inconsistent astrometric covariances



$$y^2 = (m(x) \times x)^2 + n^2$$
$$m(x) = 1 - L + L \times \left[1 + \exp \left(-k \times (\log_{10}(x) - \log_{10}(x_0)) \right) \right]^{-1}$$



Conclusions

- Cross-Matches to LSST Data Releases with macauff – Form
- Upcoming LSST:UK cross-match service macauff — let me know your thoughts/needs/hopes/dreams
 - Provide tables of cross-matches between LSST and <your favourite catalogue here!>



University
of Exeter



@Onoddil

@pm.me



.github.io



Wilson & Naylor, 2017, MNRAS, 468, 2517
Wilson & Naylor, 2018a, MNRAS, 473, 5570
Wilson & Naylor, 2018b, MNRAS, 481, 2148
Wilson, 2022, RNAAS, 6, 60
Wilson, 2023, RASTI, 2, 1

<https://github.com/macauff/macauff>
<https://github.com/macauff/birnam>



Tom J Wilson @onoddil



Science and
Technology
Facilities Council



